Topological Survey of Zeta Power Converters from Classical Designs to Emerging Architectures

Showrov Rahman [1], Md. Kawsar Hossain [2], Anindya Banik [3], Mainul Islam [4], Abdullah Al-Hysam [5*], Istiak Ahmed [6]

1. Department of Electrical and Computer Engineering, University of Rhode Island, Kingston, RI 02881, USA
2. Department of Electrical and Electronic Engineering, Ahsanullah University of Science and Technology, Dhaka 1208, Bangladesh
3. Department of Electrical Engineering, National Institute of Technology Silchar, Assam, India
4. Department of Electrical and Electronics Engineering, Islamic University of Technology, Gazipur 1704, Bangladesh
5. Division of Industrial Electrical Engineering and Automation, Faculty of Engineering, Lund University, Lund, Sweden
6. School of Engineering and Energy, Murdoch University, Perth, WA 6150, Australia

*Corresponding author: Abdullah Al-Hysam (Email:abdullah.al-hysam@iea.lth.se)

Abstract: Initially introduced as the dual single-ended primary inductor converter (SEPIC), the zeta converter has since evolved significantly from the classical direct current (DC) converter. Its ability to provide both buck and boost operation while maintaining non-inverting output voltage, continuous output current with low ripple has made it increasingly popular in a wide range of applications. The converter can also operate over a wide input voltage range, making it well suited for renewable energy systems and battery-powered applications. Today, the zeta converter is widely used in photovoltaic (PV) energy systems, electric vehicles (EVs), light-emitting diode (LED) drivers, portable electronics, and modern microgrids (MGs). Its topological evolution has extended beyond the conventional non-isolated DC–DC converter to include isolated DC–DC converters, zeta-derived alternating current (AC) to DC converters, as well as DC–AC and AC–AC converter topologies that are derived from or inspired by the zeta topology. This paper reviews the topological evolution of the zeta converter across different AC and DC power conversion systems and their applications. It also presents a systematic classification of zeta and zeta-derived converters, a comparative analysis of prominent zeta converter topologies based on component count, intended applications, power rating, and key performance metrics. For DC applications, the performance metrics includes conversion efficiency, while for AC applications, it considers efficiency, power factor, and total harmonic distortion (THD). Finally, this paper discusses the future opportunities and challenges associated with the continued development and application of the zeta converter.

Keywords: AC-DC, AC-AC, converter, DC-AC, DC-DC, dual-SEPIC, zeta

1. Introduction

The rising worldwide demand for power conversion devices has accelerated the development of advanced converter topologies used in diverse energy conversion applications including electric vehicles (EVs), battery management systems (BMS), renewable energy systems, and electric grid operations[1], [2]. Among the family of non-isolated direct current (DC) converter topologies, the zeta topology has been introduced by adopting the Greek letter "ζ", indicating the sixth topology in the family after buck, boost, buck–boost, cuk, and single-ended primary inductor converter (SEPIC) [3], [4].

The zeta converter was introduced as the dual counterpart of the SEPIC topology [4]. Both converters provide non-inverting buck-boost voltage conversion through a series energy-transfer components. However, their port-current characteristics are different: the SEPIC converter generally provides continuous input current and discontinuous output current, whereas the zeta converter provides discontinuous input current and continuous output current at the load side [5]. Zeta-derived designs can also be combined with coupled inductors, switched-capacitor cells, soft-switching methods, and advanced control strategies to improve voltage regulation, current profiles, and application adaptability in renewable-energy and microgrid systems [6], [7], [8].

In the family of non-isolated converter topologies, the earlier topologies have been extensively studied, and multiple review and survey papers are available on buck [9], [10] boost [11], [12], [13], [14], [15] buck–boost [16], cuk [17], [18], and SEPIC converters [19], [20]. In contrast, the zeta topology remains relatively less explored in the existing literature, and there is still a lack of dedicated review and survey studies focused specifically on it.

In place of dedicated survey work, the zeta topology has only been discussed in a limited capacity within several DC–DC converter review papers. For example, in [21], DC–DC converter technologies are reviewed, including conventional, bidirectional, isolated, and impedance-source converters. However, with respect to the zeta topology, only a generic overview is provided, lacking an in-depth discussion of its operating principles and characteristics.

Gorji *et al.* provide an overview of bidirectional DC-DC converter topologies, both isolated and non-isolated [22]. The study also discussed linear and non-linear control strategies and advanced modulation techniques. Nevertheless, the zeta configuration is included in the overview only as a subcategory, grouped with SEPIC and cuk topologies. It also lacks an evolutionary analysis of zeta, a multidimensional matrix framework that unifies zeta architectures across all four conversion quadrants.

Over 100 non-isolated high step-up DC-DC converters were reviewed by Soltani et al. [23], with a subset of topologies utilizing coupled inductors. To establish a baseline and highlight their limitations, the methodology begins with a systematic analysis of classical non-isolated converters, including boost, buck-boost, cuk, SEPIC, and zeta converters. However, this study only presents the zeta converter as a building element within conventional DC-DC voltage-boosting networks, leaving several clear research gaps, including zeta-derived inverter topologies and the use of zeta converters in power factor correction (PFC) applications.

The features of current and available non-isolated converter trends, such as buck-boost, SEPIC, cuk, Z-source, zeta, and hybrid DC-DC converters, are highlighted in [5]. Similar to the mentioned review works, this work provides a short discussion on the zeta topology within a broader non-isolated category. In [24], bridgeless (BL) PFC converter topologies, including BL buck-boost, BL cuk, BL SEPIC, BL zeta, and BL Luo converters, are described. Key performance parameters, including voltage gain characteristics, input power factor, total harmonic distortion (THD), semiconductor voltage and current stress, and component count, are used in the comparison. The BL zeta and BL Luo converters perform better according to the review. However, this paper focuses solely on bridgeless topologies and only on AC-DC conversion.

Ref. [25] presents the significance of bidirectional DC-DC converters in hybrid energy storage systems (HESS) and classifies the converter topologies employed in HESS applications. The study also discusses the evaluation framework used for analyzing converter performance based on several technical and operational parameters. However, the review only briefly discusses zeta-based configurations and does not provide a comprehensive evolutionary analysis of zeta-derived converter families. Furthermore, the comparative discussion of isolated and non-isolated zeta topologies in terms of voltage gain, switching stress, efficiency, ripple performance, EMI characteristics, and application suitability does not provide sufficient depth and context regarding the topology and its practical design considerations.

As discussed in the previous section, the existing review articles only discuss the zeta topology with a narrow focus while maintaining a broader scope on large families or classes of converters [21], [5]. To the best of the authors' knowledge, no dedicated survey or review work has been conducted with a specific yet broad focus solely on the zeta topology, including zeta and zeta-derived converter families. This paper addresses this literature gap by providing a systematic survey and comparative analysis of various Zeta and zeta-derived topologies. As research on zeta-derived converters continues to grow, a dedicated survey on the zeta topology can provide important research insights into the evolution of zeta-based converters, their systematic classification, diverse applications, and adaptability with other suitable converter topologies, such as SEPIC. Furthermore, a comprehensive survey on zeta converters can assist researchers in developing new and novel zeta-derived topologies tailored for specific applications. To guide future advancements in high-performance power electronic systems, a thorough analysis of current technological challenges and potential research paths is also provided.

The main contributions of the review are as follows:

- An evolutionary analysis of zeta converter topologies, highlighting the timeline of development and adoption of zeta-based topologies over the years.
- A systematic classification framework for zeta-based DC–DC, AC–DC, DC–AC, and AC–AC converter topologies.
- A comparative analysis of zeta-based DC–DC (isolated/non-isolated), AC–DC (bridgeless/bridge-based)-(isolated/non-isolated), DC–AC, and AC–AC converters based on component count, efficiency, application domain, and design focus.
- Identification of key research gaps, implementation challenges, and future research directions for next-generation power conversion systems.

2. Methodology

A literature search was conducted across several scholarly databases, including Web of Science and Scopus. Peer-reviewed publications, conference proceedings, and technical reports released between 2005 and 2025 were systematically reviewed, with the primary focus on zeta converter topologies and zeta-derived converter configurations. Effective coverage of relevant domains was achieved through the strategic use of keyword combinations. Primary search keys included "zeta converter", "modified zeta converter", "isolated zeta converter", "non-isolated zeta converter", "zeta-based DC-DC converter", "zeta-based DC-AC converter", "zeta-based AC-DC converter", "zeta-based AC-AC converter", "converter control techniques", "power quality", and "harmonic distortion". Boolean operators "AND" and "OR" were employed to extract pertinent information from the selected databases. Each study had to be relevant, technically oriented, and related to zeta converter topologies, control techniques, or performance aspects to meet the inclusion criteria. Studies on alternative converters, studies with outdated or insufficient technical information, and non-English literature without appropriate translations were excluded. The topic-driven literature search was repeated to detect publication saturation and prevent unnecessary additional searches for topics that had already been explored. After eliminating duplicates, 180 distinct literature records remained. These were then reduced to 120 studies following relevance and quality screening. The selected literature was categorized by theme and subjected to qualitative evaluation. Fig. 1 shows the overall workflow.

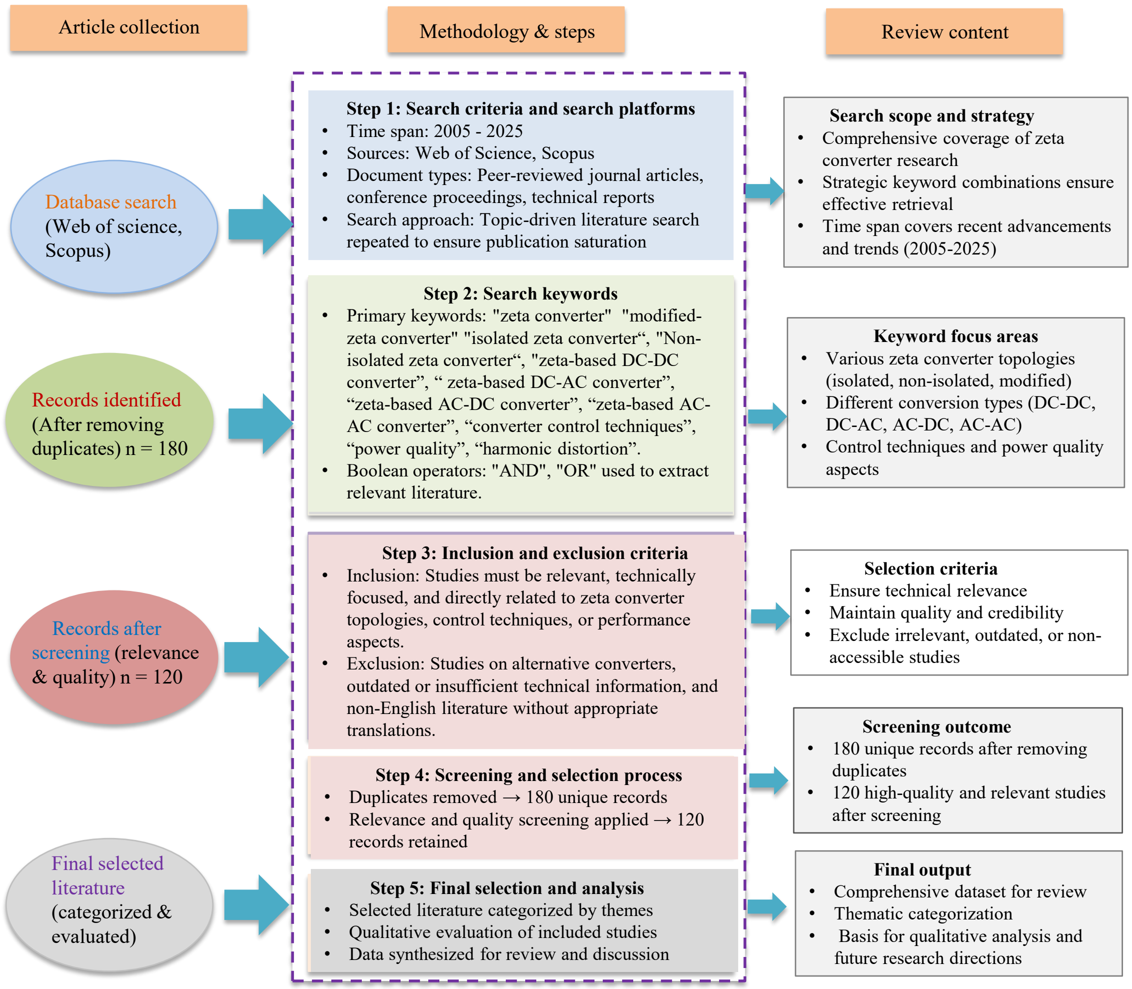


Fig. 1. Literature review, selection criteria and overall workflow.

3. Brief History and Evolution of the Zeta Topology

The zeta converter was first developed and introduced as the dual-SEPIC (single-ended primary-inductor converter) by J. J. Jozwik and M. K. Kazimierczuk in 1989 [26], [4]. The SEPIC converter, as shown in Fig. 2, is a buck-boost-mode converter capable of stepping up or stepping down the input voltage. As shown in Fig. 3, the dual-SEPIC converter is derived from the SEPIC converter by rearranging the placement of the inductors L_1 and L_2 and the active switch. Hence, the term dual-SEPIC indicates the topological duality between the two converter structures. Lengnien Hsiu et al. performed several studies on the dual-SEPIC converter with a coupled inductor in order to achieve soft-switching operation [27], [28].

According to prominent researcher and author Sam Ben-Yaakov, A. Peres et al. popularized the term "zeta" in 1994 to indicate the same converter topology previously referred to as the dual-SEPIC topology [29], [30]. However, after performing an extensive literature review, two earlier mentions of the term "zeta" converter were found before 1994: by Denizar Cruz Martins in 1991 at the 1st COBEP91 Brazilian Power Electronics Conference, Brazil, and in 1993 in the Conference Record of the Power Conversion Conference - Yokohama 1993, Japan [31].

The term "zeta" is derived from the Greek alphabet letter "zeta" (z), indicating the sixth topology in the chronological sequence after the buck, boost, buck-boost, cuk, and SEPIC converters [30]. The term "zeta" gained rapid popularity afterwards, and the topology is now widely known by this name.

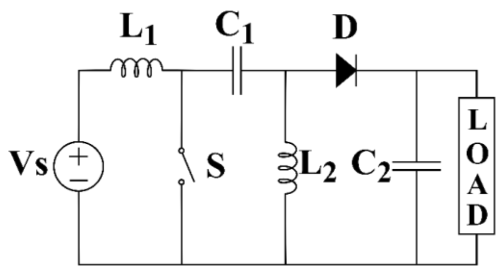


Fig. 2. Single-ended primary-inductor (SEPIC) converter.

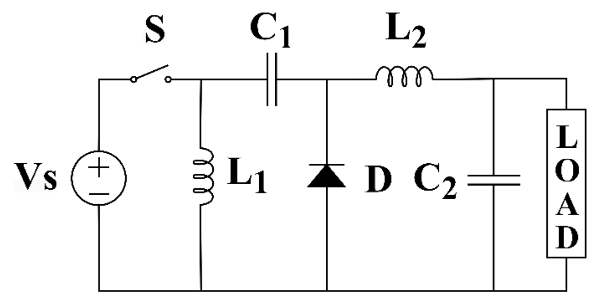


Fig. 3. Dual-Sepic converter (popularly known as zeta converter).

Both SEPIC and zeta converters have similar component counts and are capable of step-up as well as step-down operation. Both converters provide non-inverting outputs, unlike conventional buck-boost or cuk converters. Another similarity is that both topologies include a series energy-transfer capacitor. The main difference between them is the placement of the inductors and the active switch. In the SEPIC converter, the input inductor is placed in series, while the active switch S is connected in parallel. This connection is followed by a series capacitor and a parallel inductor L_2. Due to this arrangement, the SEPIC converter has continuous input current, while its output current is pulsating. In the zeta converter, the active switch is connected in series at the input side, so high-frequency input current enters the converter depending on the switching state. On the output side, the output inductor of the zeta converter helps provide continuous current to the DC load.

Both the non-isolated and isolated versions of the zeta converter were introduced by J. J. Jozwik and M. K. Kazimierczuk [26], [32], [26], [32], and in 1994 the zeta converter was utilized as a power factor correction (PFC) converter [29]. From the mid-1990s to the mid-2000s, the zeta topology gradually gained attention in various research groups. In 2005, Seong-Hwan Paeng et al. first proposed a bidirectional SEPIC/zeta converter based on the duality of the SEPIC and zeta converters. This article was published in Korean in the Proceedings of the Korean Institute of Power Electronics (KIPE) Conference. In 2007, In-Dong Kim, Seong-Hwan Paeng, and co-authors presented a similar converter topology in English at the IEEE International Symposium on Industrial Electronics, Vigo, Spain [33]. Later, the bidirectional SEPIC/zeta converter became popular for applications requiring bidirectional power flow. From around the same time, researchers started to develop various zeta-based hybrid converters, such as the zeta-flyback converter [34].

In 2005, Nam-Sup Choi proposed a three-phase direct AC-AC converter based on the zeta topology in the KIPE Conference. This article was also written in Korean. Later, in 2009, Qinglong Zhong and Lei Li presented a zeta-mode three-level AC/AC converter [35]. Starting from the early 2010s, the zeta topology started to be more commonly used in multistage cascaded DC-AC and AC-mains motor-drive conversion systems. In [36], an inverter is placed after the zeta DC converter designing zeta-derived topology. In several AC-input motor-drive systems, a zeta DC converter or zeta PFC stage is placed between an AC-DC rectifier unit and a DC-AC inverter unit [37], [38], [39]. Later, zeta-based DC-AC inverters gained popularity in PV systems, and various integrated or single-stage zeta-derived inverters were developed to replace two-stage DC-AC conversion systems in selected applications [40], [41], [42]. The evolution of the zeta converter topology is illustrated in Fig. 4 via an infographic timeline.

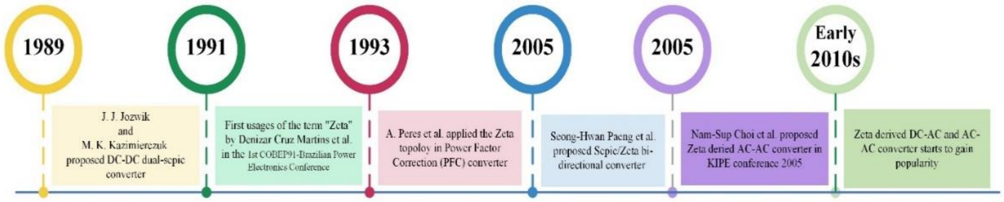


Fig 4. Info-graphic timeline showcasing the evolution of zeta topology and the major milestones.

4. Classification of Zeta Converter Topologies

Over the past years, different converters have been derived and developed from the zeta topology. Both non-isolated and isolated versions of the zeta converter were proposed by J. J. Jozwik et al. [26]. Later, the zeta topology was also employed in front-end AC-DC converters and PFC converters. In such systems, a DC-DC zeta converter or zeta-derived cell is placed after a diode bridge or bridgeless rectifier to deliver regulated DC power to the load [29], [39], [43], [44], [45], [46]. In some DC-AC applications, a zeta DC-DC converter has been employed at the input side followed by a conventional inverter circuit [47], [48], [49], [50]. These should be described as zeta-converter-fed or two-stage zeta-based inverter systems. Later, several integrated zeta-derived or zeta-based inverters were developed and used in PV applications as microinverters. The zeta topology has also been utilized in AC-AC conversion. Some AC-input systems use a zeta DC-link/PFC stage between rectification and inversion stages, while more recent literature reports direct zeta-mode AC-AC converters and zeta-based AC-link universal converters [51], [52]. These categories are distinguished in the following subsections to avoid overgeneralizing every hybrid or cascaded topology as a pure zeta converter.

Fig. 5 presents the classification of zeta-based converter topologies. The following subsections provide a concise review and discussion of the major topology categories illustrated in **Fig. 5**.

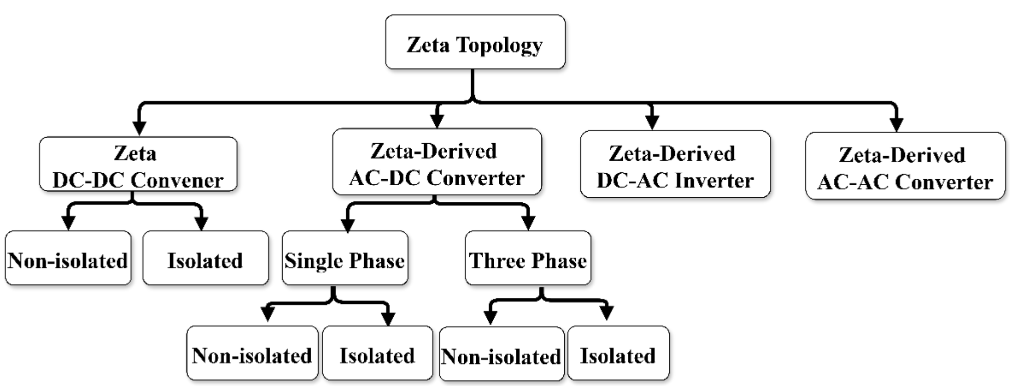


Fig 5. Classification tree diagram of zeta converter topologies.

4.1. Zeta DC-DC Converters

In this paper, we further sub-divide the DC-DC converters based on isolation type. The non-isolated and isolated zeta converters are discussed in the following subsections.

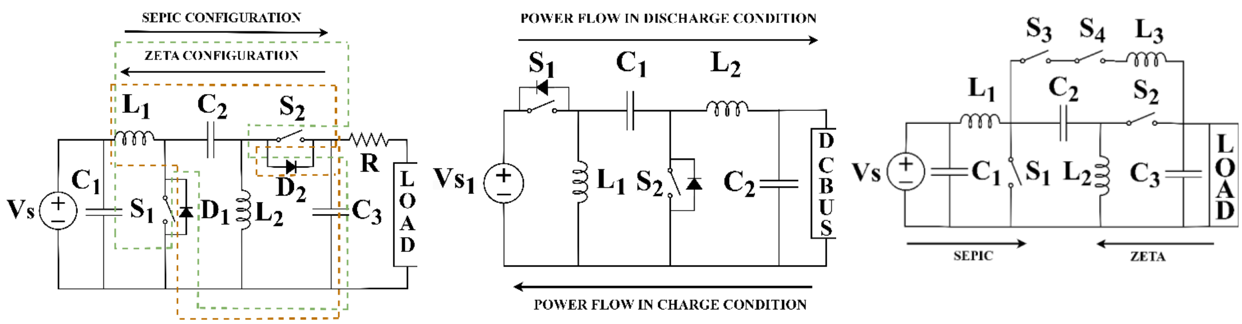
4.1.1. Non-Isolated DC-DC Zeta Converter:

The conventional non-isolated DC-DC zeta converter is shown on Fig. 6. The converter consists of two inductors, two capacitors, one active switch (e.g., MOSFET or IGBT), and one passive switch (e.g., diode). The circuit operation of the conventional DC-DC zeta converter is briefly described next. Assuming initially uncharged inductive and capacitive elements, when switch S is turned on, inductor L_1 is charged. In the next interval, switch S is turned off and the diode becomes forward biased, discharging L1 and charging the series capacitor C_1. With C_1 charged, when switch S is turned on again and diode D becomes reverse biased, L_1 is charged from the supply. Simultaneously, current flows through switch S, capacitor C1, and inductor L_2 to deliver power to the load, while L_2 is also charged. In the following off interval, the diode conducts and L_2 discharges through diode D, transferring energy to the load.

Similar to the SEPIC converter, the zeta converter contains a series capacitor C_1 that acts as an energy-transfer and DC-blocking capacitor. This capacitor should not be interpreted as providing galvanic isolation; true galvanic isolation requires a transformer or an equivalent isolated coupling element. With proper operation and control, the zeta converter can achieve soft switching, which can significantly reduce switching losses. M. J. Bonato et al. implemented a switching cell consisting of two active switches with diodes, two small resonant inductors, and one small resonant capacitor [53]. The switching cell is used in place of the active switch S of the zeta converter, and with proper pulse width modulation (PWM), the cell can achieve zero-current switching (ZCS) at turn-on and zero-current and zero-voltage switching (ZCZVS) at turn-off.

Throughout the years, non-isolated DC-DC zeta converters have been employed in diverse applications, including solar power conversion [54], light-emitting diode (LED) driving [55], and microgrid interfacing [56], [57]. In [54], a zeta converter is placed after another DC-DC converter capable of providing high voltage gain, while the zeta converter is controlled to perform maximum power point tracking (MPPT) for PV applications. In [55], an auxiliary power delivery path was implemented using an auxiliary diode and switch. The proposed converter was designed for LED backlight units and was reported to achieve 96.1% efficiency at 46.5 W output power. Hybrid zeta converters have been presented in [56], [57] for grid-interfacing applications. An interleaved zeta-cuk converter is designed for interfacing a microgrid and an electric vehicle in [56], while a non-isolated boost-zeta interleaved converter is implemented for multiport operation in [57]. The zeta converter topology is also frequently combined with other converter topologies to provide higher voltage gains [58], [59], [60], [61].

In recent years, topologies have also been extensively used in applications where bidirectional power flow is required [33], [62], [63], [64], [65], [66]. In a bidirectional converter, the source and load can interchange their roles, meaning both ports can deliver or receive power depending on the operating condition. The SEPIC/zeta topological combination is particularly useful in such applications because of the duality between the SEPIC and zeta converters. Fig. 6 displays some proposed bidirectional SEPIC/zeta converter topologies. The passive diode element in Fig. 6(a)-(b) is replaced by an active switch. Due to the connection of the inductors, the operation of the active switches, and their inherent body diodes, the converter configuration operates as a SEPIC converter in one power-flow direction and as a zeta converter in the opposite direction. In Fig. 6(b), a SEPIC/zeta converter is connected to a DC bus, and from the source to the DC bus, the converter is configured as a zeta converter. With proper control and operation, the converter can function in reverse power flow as a SEPIC converter, where the DC bus supplies power to the source. Another SEPIC/zeta combined converter was proposed by M. S. Song et al. with an additional path containing two auxiliary switches and one auxiliary inductor [62]. The proposed converter has a rated power of 320 W, allows soft-switching operation, and can reduce the inductor current ripple. The authors reported improved performance compared to the conventional SEPIC/zeta converter without auxiliary components. Later, V. Mahendran et al. worked on a similar SEPIC/zeta converter topology, implementing a 250 W prototype with field-programmable gate array (FPGA)-based control [63]. This study also shows lower inductor current ripple, and the converter prototype achieved a peak efficiency of 93% at 200 W output power.



(a) (b) (c)

Fig. 6.Bi-directional converters based on Sepic/zeta converter topologies (a)-[64] (b)-[65] (c)-[63] (The figures are redrawn and reused under the Creative Common CC-BY license).

Table 1 summarizes the key features and performance characteristics of notable non-isolated DC-DC zeta converter topologies. Table 1 lists different zeta-derived non-isolated DC-DC converter prototypes for a wide range of power levels reaching up to 1 kW [33], [56]. In the table, the symbols "D", "SW", "L", and "C" refer to diode, active switch, inductor, and capacitor, respectively. The component counts provide an approximate indication of converter size, cost, and the associated design complexity for control and thermal management. While listing the component counts, the body diodes of the active switches are not counted because they are integrated parts of the active switches. Likewise, parasitic elements such as parasitic switch capacitance, leakage inductance, and line inductance are not counted. Filtering elements, such as filter capacitors and/or inductors, are included in the comparative analysis because they are externally added and are essential parts of the converters. The converter operating modes, rated power, and efficiency are also listed in the table.

Table 1. Evaluation of recent non-isolated DC-DC zeta converter-based studies [D: diode; SW: switch; L: inductor; C: capacitor]

Year	**D**	**SW**	**L**	**C**	**Mode**	**Rated level**	**Efficiency**	**Specific Applications**	**Study Focus**	**Ref.**
1989	1	1	2	2	CCM & DCM	8.91 W @CCM, output voltage 9.44 V 4.98 W @DCM, output	Not reported	General purpose	Steady-state analysis and experimental evaluation of dual-SEPIC (zeta) converter	[26]

Year	D	SW	L	C	Mode	Rated level	Efficiency	Specific Applications	Study Focus	Ref.
						voltage 14.12 V				
2000	2	2	4	3	Not reported	500 W	97.3% @500 W, output voltage 110V, input voltage 220V	Both DC-DC and AC-DC conversion system	Switching cell for zeta DC-DC converter for high efficiency soft-switching operation	[53]
2007	4	4	3	4	Not reported	1000 W	~88% @ 700W output power	Application requiring Bi-directional power flow	Design and experimental analysis of bidirectional ZVS PWM Sepic/zeta DC-DC converter	[33]
2011	2	2	2	3	CCM	46.5 W @ output voltage 24 V; output current 1.9 A	96.1% @ 46.5 W, output voltage 24 V, output current 1.9 A	Direct-lit light emitting diode (LED) backlight unit.	Implementing additional power delivery path using a diode and switch combined with a conventional zeta converter	[55]
2014	0	4	3	3	CCM	320 W	≈94.4% @zeta operation; output power 320 W	Application requiring Bi-directional power flow	Bidirectional Sepic/zeta DC-DC converter with soft-switching operation	[62]
2015	9	6	2	6	Not reported	100 W @ output voltage 220 V	91% @ 100 W	PV Applications	Design and experimental analysis of extended zeta converter for PV applications	[54]
2017	0	4	3	3	CCM	250 W	93% @200W load	Application requiring Bi-directional power flow	Analysis and practical implementation of a bi-directional Sepic/ zeta converter	[63]
2018	0	2	2	3	CCM	12 Volt, 1.6 Ampere-hour (Ah) lead-acid battery cell	Not reported	Application requiring Bi-directional power flow	Dynamic performance study focusing bidirectional Sepic/zeta feature	[64]
2019	1	7	2	4	CCM	Not reported	~98% @ 4kW output power (analytical)	General purpose	Mathematical models and experimental verification of switched capacitor-based zeta converter	[67]
2020	2	1	3	4	CCM & DCM	200 W @ output voltage 80 V	~97% @ 25 W	General purpose	Mathematical models and experimental verification of a new buck–boost converter based on a ZETA topology	[68]
2020	2	1	2	3	CCM	24 W	Not reported	General purpose	Investigation of zeta-buck-boost converter controlled by a non-linear controller.	[69]
2020	1	1	2	2	DCM	325 W	Not reported	Brushless direct current (BLDC) motor driver	Notch filter modeling for reducing commutation current ripple in zeta converter	[70]
2020	0	2	2	2	Not reported	Not reported	92.5% @12.8 V Energy storage, 18 V DC bus voltage	Application requiring Bi-directional power flow	Bidirectional zeta/Sepic converter between energy storage device and DC bus system	[65]
2021	2	1	3	4	CCM, DCM	300W	94.3% @ 240V output voltage	Application requiring wide voltage conversion ratios	Operation, design and analysis of zeta converter by incorporating a coat circuit for CCM and DCM mode operation.	[59]

Year	D	SW	L	C	Mode	Rated level	Efficiency	Specific Applications	Study Focus	Ref.
							94.2% @ 240W output power			
2021	8	2	5	3	CCM & DCM	500 W	92.93% @ 500 W	General purpose	Steady-state analysis, non-ideal model and experimental evaluation of hybrid converter based on zeta and boost converter, with active quad switched-inductor	[60]
2022	0	2	2	2	CCM	18.35 W @ supercapacitor string charging 29.84 W @ battery string charging	91.2% @ supercapacitor string charging 91.7% @ battery string charging	Application requiring Bi-directional power flow	Bidirectional energy transfer between supercapacitor string and battery string. Automatic equalization of the hybrid energy storage system; assessment, and experimental validation.	[66]
2022	2	2	2	2	CCM	173 W	95.2%	General purpose	Steady-state and dynamic analyses of a hybrid zeta	[61]
2023	1	1	3	3	CCM	100W	> 96% @ entire load range	General purpose	Soft-switched couple inductors based step-down zeta converter for low output current ripple	[71]
2023	2	2	4	3	CCM	1000W	97.8%	Microgrid and electric vehicle (EV)	Analysis of a novel converter based on cuk-zeta combination.	[56]
2023	3	2	4	4	CCM, DCM	400W	95.8% @full load	General purpose	Operating principle, steady-state analysis, CCM-DCM boundary analysis, analytical modeling of zeta-buck-boost integrated converter with ZCS turn-on.	[72]
2023	2	2	3	3	CCM & DCM	192W prototype @64W positive pole, 128W negative pole	94.42% @full load	Low-voltage DC microgrid (monopolar and bipolar)	Steady-state and dynamic analysis, CCM-DCM boundary analysis of a novel boost-zeta non-isolated interleaved converter.	[57]
2023	1	1	2	3	CCM	Vin: 400V; Vo: 360V; Power Not Reported	95.19% for dual Load @Vo=360V	Solar PV energy system	Developed Hybrid PSO-GWO optimization for passive component selection.	[73]
2024	0	4	3	3	Not Reported	Not Reported	96%	Vehicle to Grid (V2G) integration	modeling of a bidirectional SEPIC-zeta converter controlled by ANFIS and CNN based SOC estimation	[74]
2024	1	1	2	3	CCM	120w @voltage level 594V/60V	91.8%	Electric Vehicle (EV) battery charging stations	closed loop battery current-controlled zeta converter, power and frequency analysis of charging station.	[75]
2024	4	2	3	6	CCM, DCM	400w @voltage level 32/400V	95.9% full-load efficiency @ 400 W, 400 V	PV generation systems	Steady-state analysis, topology variation and experimental validation of interleaved Buck-boost-zeta converter with zero input current ripple.	[76]

Year	D	SW	L	C	Mode	Rated level	Efficiency	Specific Applications	Study Focus	Ref.
2024	2	2	4	4	Not Reported	10Kw @ voltage level 57-69V/600V	97.9%	Hybrid Renewable Energy System (HRES) based microgrid	modeling and control of a microgrid with novel zeta converter using GWOSL-optimized PI controller.	[77]
2024	1 2	2	2	5	CCM	Vin: 15V; Vo: 12-16V(battery string), 0-28.8V(super capacitor string); Power Not Reported	86.4%	HESS for electric vehicles	Steady-state analysis, charging mode analysis, experimental validation of a zeta converter based charging equalizer.	[78]
2024	3	1	1	3	CCM, DCM	160w @ voltage Level 36-48V/220V	94.8% @ 160 W- 36 V input 95.5% @ 160 W- 48 V input	New energy power generation systems	Steady-state analysis, CCM-DCM boundary analysis, small-signal modeling and experimental validation of coupled inductor boost-zeta converter.	[79]
2024	3	2	3	4	CCM, DCM	Buck: 40w @ 20/10V. boost: 60w @20/75V	boost Mode: 97% @ 100 W output power (Based On analysis) Buck Mode: 92% @ 100 W output power (Based on analysis)	For applications in renewable energy systems and industrial settings	Steady-state analysis, parasitic efficiency calculation, small-signal modeling and experimental validation of a semi-quadratic buck-boost converter.	[80]
2025	3	6	6	4	CCM	6000w @ voltage level 480V/250V	98.16% @6kw	Application requiring Renewable-powered electrolyzers (PEMEC)	Steady State analysis in all three operating regions, current ripple analysis and experimental validation of Interleaved zeta Converter.	[81]
2025	1	1	2	2	CCM	Vin: 38V , Vo: 13V; Power Not Reported.	Not Reported	PV Systems	GA-based PID controller optimization for a zeta converter and stability analysis.	[82]
2025	0	2	2	2	CCM	80w @voltage level 9-14V/24V.	Not Reported	General Purpose	Mathematical modelling and implementation of a feedback Linearization to synchronous zeta Converter.	[83]
2025	4	2	4	4	DCM	DC Output: 360V; Power Not Reported	Not Reported	Air Conditioning System (ACS)	Performance assessment of bridgeless zeta converter and solar energized zeta converter for ACS	[84]
2026	1	1	2	2	Not Reported	Not Reported	99%	Electric Vehicle	Analysis of a zeta converter with MOA optimized FOPID controller for EV applications and experimental validation of the system.	[85]

4.1.2. Isolated DC-DC Zeta Converter:

A comparative analysis of notable isolated DC-DC zeta converters is listed in Table 2. One of the key elements of an isolated converter is the isolation transformer, which can be designed in various ways. In Table 2, transformers are counted based on the number of transformer cores. For example, an isolated DC-DC zeta converter was introduced in [26], where an isolation transformer with a single primary winding and a single secondary winding is used. On the other hand, a multi-output isolation transformer can employ several secondary windings and a single primary winding while still using a single transformer core. In both cases, the transformer is counted as a single transformer because a single magnetic core is employed. For multiple-winding transformers, the winding numbers are listed in the table for easier identification [31], [34], [86], [87].

The zeta-derived single-winding (single-input single-output, SISO) and multiple-winding (single-input multiple-output, SIMO) isolated DC-DC converters are displayed in Fig. 7.

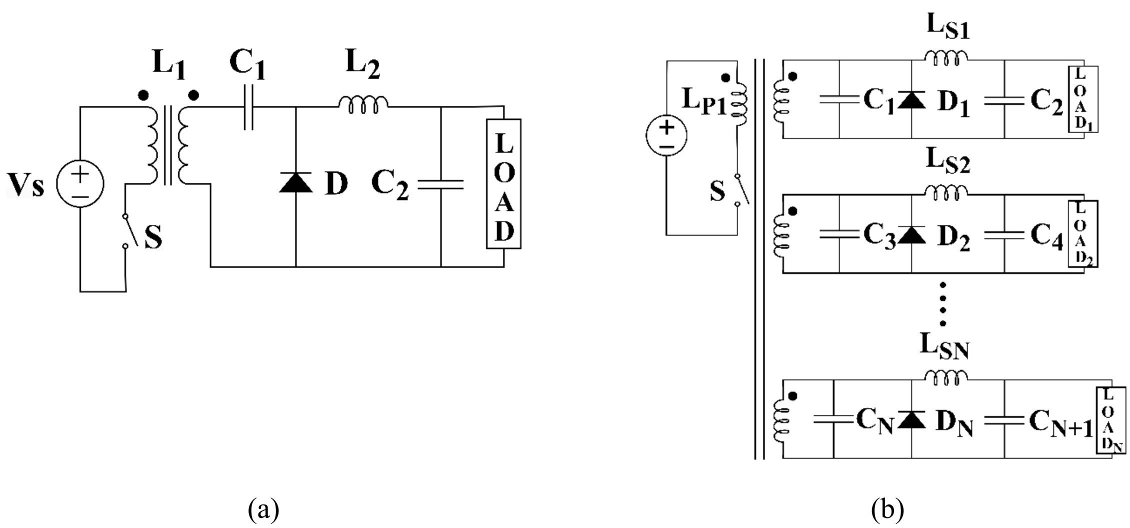


Fig 7. (a)-zeta derived single winding (single input single output- SISO); (b)- zeta derived multiple winding (single input multiple output- SIMO)

As previously mentioned, the table emphasizes studies that provide experimental validation where available; however, a small number of simulation-based studies are retained when they are directly relevant to the zeta-derived isolated converter family. While listing the component counts, the body diodes of active switches are not counted because the body diode is an integrated part of an active switch. The parasitic elements of the transformer, such as leakage inductance, interwinding capacitance, and resistance, are also not counted. Similarly, the parasitic elements of active switches are excluded from the component counts. Filtering elements, such as filter capacitors and/or inductors, are included in the comparative analysis because they are externally added and essential to converter operation. The converter operating modes, rated power, and efficiency are also listed in Table 2.

Over the years, various modifications have been performed on the basic zeta converter to improve its efficiency and reduce device stress. In [88] and [87], it has been shown that the inclusion of a clamping circuit can reduce the voltage stress of switches. Active clamping also helps limit the maximum voltage stress of transistors [89]. It has been used in different proposed converters mentioned in [88], [90], [91], [92], [93].

Jae-Bum Lee et al. proposed a zeta converter with an asymmetrical half bridge (AHB) that also achieves a wide zero-voltage-switching range with a small transformer leakage inductor [86]. Jaeil Baek et al. presented a semi-resonant half-bridge zeta converter capable of achieving a wide zero-voltage-switching range and suitable for high input-output voltage amplification [87]. Ki-Bum Park et al. proposed a double-ended ZVS half-bridge zeta converter (DHBZ), which shows high efficiency over a wide range of load variation [94]. The proposed converter takes DC input and performs high-frequency DC-AC conversion followed by transformer isolation and secondary-side AC-DC rectification. Thus, it can be considered an isolated DC-DC zeta-derived converter, as shown in Fig. 9. In [87], a bidirectional zeta converter was proposed for applications requiring a high voltage conversion ratio.

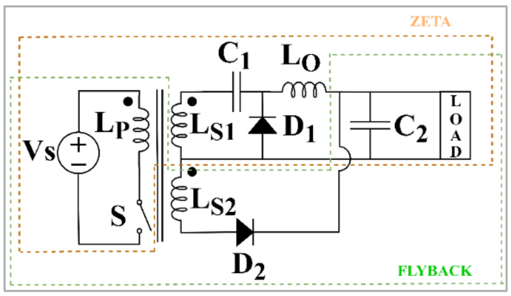


Fig 8. Center Taped zeta Flyback Converter [88]

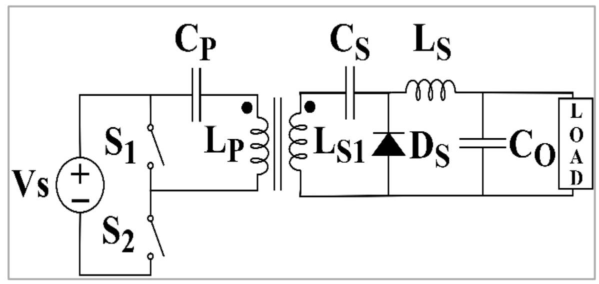


Fig 9. Two stage Isolated DC-DC zeta Converter (Half Bridge) [94]

For specialized applications, specific converter topologies can be introduced by modifying or hybridizing the zeta topology. Bor-Ren Lin et al. implemented a soft-switching zeta-flyback converter that provides lower output voltage fluctuation under variable load conditions [88]. The proposed converter is a hybrid topology where a center-tapped isolated structure is used. At the secondary side, one section of the converter acts as a zeta converter and the other portion acts as a flyback converter, as displayed in **Fig. 8**. Different modified zeta converters are used in various applications, including external power supplies (EPS) [87], uninterruptible power supplies (UPS), energy storage [92], and LED lighting [93]. Zeta converters proposed in [87] and [94] are suitable for cases where high output voltage is required.

Table 2. Evaluation of recent isolated DC-DC zeta converter-based studies [D: diode; SW: switch; L: inductor; C: capacitor; TR: transformer; CT: center tapped transformer]

Year	D	SW	L	TR	C	Mode	Rated Power	Efficiency	Specific Applications	Study Focus	Ref.
1993	4	1	4	1 [4 sec. winding]	8	CCM	200 W	~80% @ full load 200 W	General purpose	Desing and practical implementation of multi-output (four secondary winding) isolated zeta converter	[31]
2003	1	2	1	1	2	CCM	60 W	91.23% @ 60 W	General purpose	Design, derivation, and implementation of an asymmetrical PWM controlled isolated DC-DC zeta converter.	[111]
2007	2	2	2	1 [2 sec. winding; CT]	2	CCM	240 W	90.5% @ 240 W	General purpose	Steady-state analysis, ZVS principle, and experimental validation	[34]

Year	D	SW	L	TR	C	Mode	Rated Power	Efficiency	Specific Applications	Study Focus	Ref.
2008	2	2	1	1	4	CCM	400 W	94% @with RC snubber 96% @with lossless snubber	High output voltage applications	Zeta converter with a double-ended half-bridge topology containing snubber circuit for soft-switching and reduced component stress	[115]
2009	3	2	3	2	4	Not reported	360 W	89.3% @ rated output power - 360W	General purpose	Integration of flyback and zeta converter with active clamping technique for zero voltage turn on	[112]
2011	3	2	2	1	2	CCM	30 W	86% @ 30 W 88% @22.5 W	General purpose	Steady-state analysis and experimental validation	[110]
2011	1	2	1	1	3	Not Reported	180 W	~94.5% @60% Load	General Purpose	Design and experimental validation of a ZVS half-bridge zeta converter with center-tapped rectifier	[95]
2014	0	4	1	1	5	Not reported	500 W	94.3% @150W; discharge mode 92.6 % @100 W; charge mode	Application requiring Bi-directional power flow	Design and experimental implementation of a bidirectional converter based on zeta topology	[96]
2015	1	2	1	1 [2 sec. winding; CT]	4	Not reported	250 W	~93.5% @ 50% load; 400 V input voltage	Wide input voltage and low output current applications	Design and implementation of asymmetrical half-bridge converter employing a novel center-tapped zeta rectifier	[108]
2015	2	3	1	1	3	Not Reported	90-180 W	95.5% @40 V output	LED driver	Integration of AHB and HBZ converters with snubber switch	[97]
2017	1	2	2	1	3	Not Reported	250 W	94.7% @ 50% Load	General Purpose	Experimental validation of a resonant half-bridge zeta converter for reduced peak current	[98]
2020	1	2	1	1 [2 sec. winding; CT]	4	Not reported	250 W	94.77% @ 50 % load	High efficiency external power supply (EPS)	Highly efficient semi-resonant half-bridge zeta converter modeling	[99]
2022	0	2 + 1(relay)	0	1	3	CCM	50 W	87% @ full load-50 W	Solar / PV Application	Analysis of isolated zeta for battery charging, incorporation of flyback to feed the load by discharging battery	[114]
2022	5	3	2	1	3	CCM	25 W	Not Reported	UPS, EVs, Renewable Energy Applications	State-space modeling and simulation of a triple transistor isolated bidirectional zeta-SEPIC DC-DC converter	[100]
2025	3	1	2	1	5	CCM	150 W	91.59% @30V Input voltage	EV charging	Design and experimental validation of dual-mode hybrid isolated zeta-boost converter	[101]

4.2. Zeta-Based AC-DC Converters

The zeta-based AC-DC converters discussed in this paper mainly indicate converters where zeta topology is used after the rectification stage. Based on the isolation approach used after the rectification stage, the converters are further sub-classified as non-isolated and isolated AC-DC zeta converters.

4.2.1. Non-Isolated AC-DC Zeta Converter:

Recently, non-isolated AC-DC zeta topologies have become popular for developing power factor correction (PFC) converters [102], [103]. Most of the proposed zeta converters are based on passive rectification and use a full-bridge rectifier at the input side [103], [53], [104], [105], [106], [107]. The schematic diagram of a single-phase full-bridge AC-DC zeta rectifier is shown in Fig. 10(a). This converter typically uses a diode bridge rectifier containing four diodes, which converts the AC supply to DC. The second stage is a DC-DC zeta converter that supplies regulated DC power to the load. By combining the AC-DC rectifier and DC-DC zeta stages, this zeta-based converter can operate from AC supply sources and support DC loads with variable voltage regulation.

In some literature, zeta converters are also combined with bridgeless rectifier topologies. For example, researchers from the Indian Institute of Technology Delhi (IIT Delhi) developed a bridgeless non-isolated AC-DC zeta converter. The developed bridgeless converter is shown in Fig. 10(b). The applicability of the converter has been tested for motor driving [38], and later similar AC-DC zeta converter topologies were also used in LED lighting applications [108]. Both application fields are increasingly using zeta-derived AC-DC converters [109], [105]. For example, Somnath Pal et al. developed a converter achieving nearly unity power factor and minimal harmonic distortion, applied in high-power LED lighting [105]. Shyam Sunder Sharma et al. established the design of a zeta converter for dimming LEDs using two sensors [106]. B. Singh et al. performed an experiment for improving power quality in permanent magnet synchronous motor (PMSM) drives using a proposed zeta converter [104]. M. El-Shebiny et al. developed a zeta converter to control the speed of a DC motor using a single switch [110]. AC-DC zeta converters are also being used in EV applications. For example, A. K. Singh et al. presented a single-stage power converter for plug-in EVs that supports a wide input-voltage range [107]. A detailed comparison of non-isolated AC-DC zeta converters is listed in Table 3.

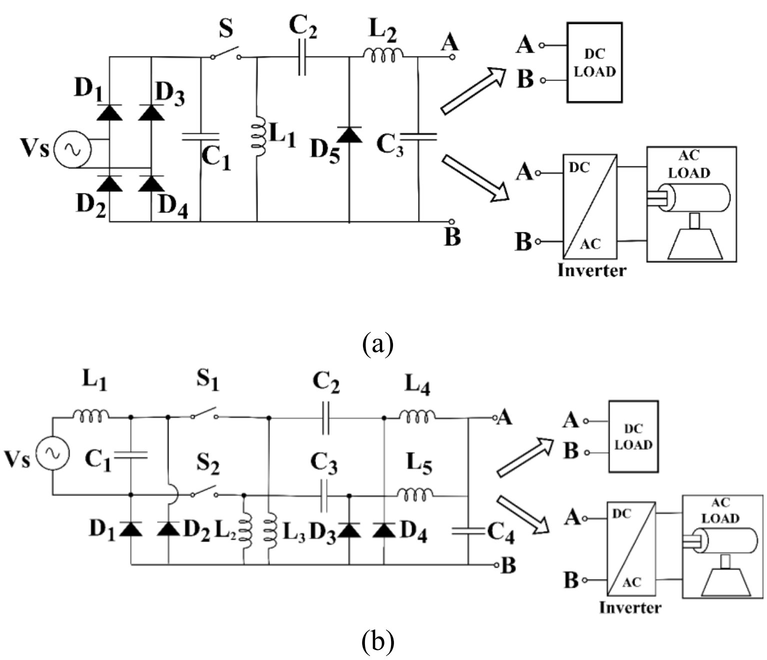


Fig. 10. (a)- Zeta converter with full bridge rectifier; (b)- Bridgeless Zeta Converter [8].

Table 3. Evaluation of recent non-isolated AC-DC zeta converter-based studies [D: diode; SW: switch; L: inductor; C: capacitor, THD: total harmonic distortion, THDi: total harmonic distortion of converter current]

Year	Rectification type	D	SW	L	C	Mode	Rated Power	Power Factor	THD	Efficiency	Applications	Study Focus	Ref.
2006	Diode bridge	5	1	3	3	DCM	2500 W	0.998 (@ full load)	THDi (input current) 6.13% for 20% load (400W), 3.85% for full load 2000 Watt	Not Reported	PMSM drive	DCM analysis and experimental validation of zeta in the context of VSI (voltage source inverter)-fed PMSM drive. The formulation of the control scheme of the VSI-fed PMSM drive was also thoroughly investigated.	[111]
2008	Diode bridge	5	1	2	2	N/R	N/R	~0.952 (@ 0.5 full load)	N/A	Not Reported	DC motor drive	Mathematical models and experimental verification of speed control of DC motor using zeta converter	[110]
2012	Diode bridge	5	1	2	2	CCM	75 W	0.996 @normal operation ***** 0.877 @distortion operation	THDi: 9.35% @normal operation ***** THDi 54.7% @distortion operation	94% @75 W	General purpose application	Investigating distortion behavior of zeta converter	[102]
2012	Bridgeless	6	1	6	4	Not Reported	Not Reported	1 @D= 0.3	3.47% @D= 0.3	95.42% @D=0.3	General purpose	Design and analysis of Single-switch bridgeless AC-DC zeta converter	[112]
2013	Diode bridge	6	2	5	5	DCM	400 W @ Vo = 200 V	~0.99 @ full load	THDi (input current) <10%	~90.8% @ full load and Vin = 110 V	General purpose application	Steady state analysis and experimental validation of an Interleaved Single-Phase PFC zeta Converter	[103]
2013	Bridgeless	4	2	5	4	DCM	800 W	~0.99 @ wide load range	THDi 2.12% @rated load	Not Reported	Brushless DC motor driver (BLDC)	DCM-operated bridgeless zeta converter for BLDC drive.	[38]

Year	Rectification type	D	SW	L	C	Mode	Rated Power	Power Factor	THD	Efficiency	Applications	Study Focus	Ref.
2014	Diode bridge	5	1	3	3	DCM	350 Watt	1 (@317 input ac power)	THDi 4% (input current)	Not Reported	Brushless DC (BLDC) motor	ZETA PFC converter to supply high quality power to a voltage source inverter to drive BLDC motor	[109]
2017	Bridgeless	4	2	4	3	DCM	300 W	1 @ full load	THDi 2.9% @ full load	~84% @ 67% load (overall system efficiency)	Low voltage high current (LVHC) LED driver	Design and implementation of PFC based Bridgeless (BL) ZETA LED driver	[108]
2018	Diode bridge	4	3	3	3	CCM	215 W	0.99 @ 226 W, 100 V peak grid voltage	THDi 4.8% @ 3.21 Amp, 210 W charging, 100 V (peak) grid voltage	94.76% @ 85W, 60 V peak grid voltage	Plugin EV	Design, analysis, and control of multifunctional zeta-SEPIC converter in context of EV applications.	[107]
2021	Diode bridge	5	1	3	3	CrCM	150 W	> 0.98 @265V ac input ac voltage	THDi 5.6% @265 Vac	94%	LED lighting applications	Sizing guidelines, steady-state, and dynamic analyses of flyback zeta converter. Modeling of a LLC resonant converter cascaded with flyback zeta PFC for DC-DC power stage.	[105]
2021	Bridgeless	4	2	5	4	DCM	100 W	0.9990 @120V load	2.13@120V load	Not Reported	Switched Reluctance Motor (SRM) drive	Design and validation of Single-stage bridgeless DCM zeta converter with PFC for SRM drive	[113]
2022	Diode Bridge	8	1	2	4	DICM	750 W	1 @230V	3.89% @230V	Not Reported	Low Voltage Electric Vehicles	Design of modified zeta converter with switched capacitor and coupled inductor for high step-down gain	[114]
2023	Diode Bridge	5	1	4	3	DCM (input side), CCM	28.35 W @ Vo = 81V, Io = 350mA	Not Reported	THDi 17.5% @ LED Power: 27 W	Not Reported	LED driver	Design and experimental verification of zeta Converter-Based	[106]

Year	Rectification type	D	SW	L	C	Mode	Rated Power	Power Factor	THD	Efficiency	Applications	Study Focus	Ref.
						(load side)			(100% dimming) ******* THDi 10.3% @ LED Power: 14 W (50% dimming)			Dimmable LED Light System	
2025	Bridgeless	3	4	4	4	DICM	400 W	0.9986	3.07%	94.2%	On-board EV charger	Design and validation of bridgeless zeta-derived PFC converter with reduced magnetic components.	[115]
2025	Diode Bridge	6	2	5	4	DCM	1 kW	Not Reported	<2%	Not Reported	EV off-board chargers	Design and analysis of interleaved zeta converter with DCM operation	[116]
2025	Bridgeless	4	2	5	4	DICM	400 W	0.9983	3.46%	94.38%	Battery charging, portable DC power supplies	Performance analysis of multi-port bridgeless zeta converter for high power quality and reduced conduction losses.	[117]
2025	Bridgeless	2	2	3	4	DCM	1 kW	1	3.39%	93.08%	On-board EV charger	Design and validation of bridgeless zeta PFC converter with reduced DC-link capacitance	[118]

4.2.2. Isolated Zeta-Based AC-DC/PFC Converter:

Isolated zeta-based AC-DC converters are rectifier/PFC systems in which the regulated isolated power stage is derived from the conventional DC-DC zeta converter. In these converters, a high-frequency transformer provides galvanic isolation and may replace or integrate the input-side magnetic energy-transfer function of the non-isolated zeta cell. This modification maintains the buck-boost functionality and continuous output-current characteristic associated with the zeta topology. The input stage typically uses a diode bridge rectifier, as shown in Fig. 11, although a bridgeless configuration can also be employed to reduce conduction losses [119]. An LC filter is commonly used at the input side to reduce switching ripple from the supply mains. The converter is capable of achieving a good power factor, especially while operating in DCM, where the input current can be shaped to follow the input voltage waveform. For comparatively high-power applications, continuous conduction mode (CCM) operation can also be adopted by using more complex control strategies [120]. Over time, various modification methods have been introduced to optimize the operation of isolated AC-DC zeta converters. By adding a non-dissipative snubber, the energy associated with leakage inductance can be recovered, thereby increasing overall system efficiency [121]. A two-switch configuration has also been demonstrated to keep switch voltage stress clamped to the input DC level [122]. In addition, multiple secondary windings can be integrated on the same transformer core to generate two isolated secondary voltages without the use of a separate transformer core [123]. These developments show the flexibility of isolated zeta-based AC-DC converters for applications that require both improved power quality and galvanic isolation.

Selected isolated zeta-based AC-DC/PFC converters are compared in Table 4. The symbols "D", "SW", "L", "C", "TR", and "THD" stand for diode, active switch, inductor, capacitor, isolation transformer, and total harmonic distortion, respectively. The number of components provides an indication of converter size, implementation cost, and the complexity associated with control and thermal management. In this comparison, the body diodes of active switches and the parasitic elements of transformers and semiconductor devices are not included in the component count. However, externally added components, such as input LC filters and snubber circuits, are considered because they play essential roles in converter operation. The number of transformers depends on the number of magnetic cores of the converter; a single-winding transformer is considered one transformer, and a multi-winding transformer with more than one secondary winding sharing the same magnetic core is treated as a single transformer unit.

The majority of the converters reviewed employ a diode-bridge rectifier in the input stage, while a smaller number of studies adopt bridgeless structures to reduce conduction losses. Single-switch topologies are more frequent due to their simpler circuit structure and control implementation, while two-switch topologies are mainly reported for higher power-handling capability or reduced voltage stress. DCM is the dominant operating mode in the reported works.

The power ratings vary from low power, such as 30 W [124] and 90 W [125], to medium power between 250 W and 1 kW, with one study [126] mentioning a 60-kW motor-drive system. Most of the converters achieve a power factor close to unity along with low THD, indicating satisfactory input current quality. Depending on the converter structure and operating conditions, the efficiency ranges from 75% [44] to approximately 95% [125]. The main application areas are LED driver circuits, SRM, BLDC and PMBLDC motor drives, electric vehicle battery charging systems, and general-purpose power supplies.

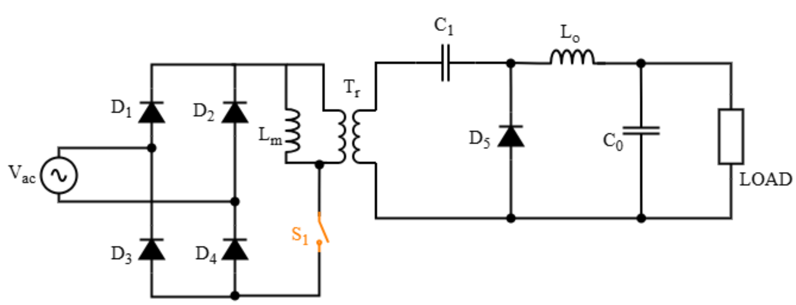


Fig. 11: Circuit diagram of an isolated Zeta AC-DC converter.

Table 4. Evaluation of recent isolated AC-DC zeta converter-based studies [D: diode; SW: switch; L: inductor; C: capacitor; TR: transformer; CT: center tapped transformer]

Year	Rectification type	D	SW	L	TR	C	Mode	Rated Power	Power Factor	THD	Efficiency	Applications	Study Focus	Ref.
2005	Diode Bridge	5	1	1	1	2	DCM	36 W	0.987	9%	75%	General purpose	Small-signal modeling and validation of isolated zeta converter	[44]
2006	Diode Bridge	5	1	2	1	3	DCM	250 W	0.998 @full load	6% @full load	Not Reported	General purpose	Performance comparison of four high-frequency isolated AC-DC converters	[127]
2010	Diode Bridge	5	1	2	1	2	CCM	800 W @RL loads	0.8842	28.83%	Not Reported	General purpose	steady-state and small-signal modeling, transfer function derivation of isolated CCM zeta converter	[128]
2013	Diode Bridge	7	1	3	1	5	DCM	400 W	1 @67.3 % laod	Not Reported	92%	Telecommunication power supplies, domestic electric appliances, battery charges, UPS	Design and validation of isolated zeta converter with non-dissipative snubber	[164]
2015	Diode Bridge	7	2	2	1	3	DCM	760 W	~ 0.99	3.6% @rated load	90.6%	Arc welding power supply	Design validation of a two-switch modified isolated zeta converter	[165]
2016	Diode Bridge	5	1	2	1	3	DCM	60 kW (motor rating)	~1	7.68%	Not Reported	Switched Reluctance Motor drive	Design of an Isolated zeta converter in DCM for PFC	[169]
2016	Bridgeless	2	2	3	2	6	DCM	90 W	0.9954	Not Reported	94.98%	LED Lamp Driver	Analysis of Fuzzy Logic Controller based bridgeless isolated interleaved zeta converter	[129]
2016	Bridgeless	3	2	2	1	3	DCM	1 kW	1	1.54%	Not Reported	EV battery charger	Design and simulation of bridgeless zeta converter with CC-CV control	[130]
2018	Diode Bridge	6	1	3	1	5	DCM	200 W	1	3.9% @90V input	Not Reported	Switched Reluctance Motor drive	Analysis of Isolated dual-output zeta converter with PFC	[166]
2019	Diode Bridge	5	1	1	1	2	DCM	36 W	0.97 @270V	7.54% @270V	Not Reported	LED lighting	Design and analysis of DCM operated single-stage isolated zeta	[131]

Year	Rectification type	D	SW	L	TR	C	Mode	Rated Power	Power Factor	THD	Efficiency	Applications	Study Focus	Ref.
													converter for PFC and low THD	
2020	Diode Bridge	5	1	2	1	3	DCM	Not Reported	0.997 @120V input	3.3% @120V input	90.8% @120V input	LED drives	Design and simulation of isolated zeta converter with PI controller for LED drive	[132]
2020	Diode Bridge	5	1	2	1	3	DCM	30 W	0.998 @340 rpm	4.54% @340 rpm	Not reported	PMBLDC Motor	Design and simulation of isolated PFC zeta converter for PMBLDC motor with variable speed control	[167]
2021	Bridgeless	4	2	2	2	4	DCM	780 W	~ 0.99	4.1%	~ 91.2%	EV battery charger	Design and experimental validation of bridgeless isolated zeta-Luo converter in DCM	[162]
2021	Bridgeless	3	2	2	1	3	DCM	Not Reported	~ 0.99	1.3%	Not Reported	battery charging for Light EV	Design and simulation of integrated bridgeless isolated zeta converter and solar fed CUK converter for battery charging	[133]
2023	Bridgeless	4	2	2	2	4	DCM	Not Reported	Not Reported	Not Reported	Not Reported	EV Charging	Design and simulation of isolated zeta-Luo converter foe EV charging	[134]
2025	Diode Bridge	5	1	2	1	3	Not Reported	840 W	Not Reported	Not Reported	Not Reported	2-wheeler EV battery charger	Design and analysis of an isolated zeta converter with open-loop and closed-loop	[135]

4.3. Zeta-Derived and Zeta-Based DC-AC Inverters

Zeta-derived and zeta-based DC-AC inverters are inverter topologies whose high-frequency energy-transfer network is derived from the conventional DC-DC zeta converter. In integrated or single-stage configurations, the zeta cell is reconfigured or combined with unfolding, differential, H-bridge, bridgeless, canonical switching cell, or hybrid structures to generate an AC output while retaining buck-boost voltage capability [136], [137]. Therefore, this section uses the terms "zeta-derived" and "zeta-based" broadly. Cascaded systems consisting of a zeta DC-DC converter followed by a conventional voltage-source inverter should instead be treated as zeta-converter-fed or two-stage zeta-based inverter systems. The topology can be operated in both CCM and DCM, where DCM operation provides a more linear static gain that simplifies controller design, while CCM operation can provide improved efficiency and lower RMS current stress in semiconductor devices. In the conventional non-isolated configuration, an important advantage is that some common-ground zeta-derived inverter structures can maintain a common ground between the input and output ports, effectively reducing or eliminating common-mode leakage currents in PV applications [138]. The converter family also supports buck-boost functionality, allowing the AC output voltage magnitude to be either lower or higher than the input voltage. Isolated configurations can be achieved by adding a high-frequency transformer to provide galvanic isolation [139], [140]. These characteristics make zeta-derived DC-AC converters suitable alternatives to traditional two-stage DC-AC conversion systems in selected applications.

Based on the reviewed literature, selected zeta-derived and zeta-based DC-AC inverter studies are presented in Table 5. The table includes the number of diodes, active switches, inductors, capacitors, transformers, power level, THD, efficiency, application area, and study focus. These parameters provide an indication of the physical size of the converter, implementation cost, power-handling capability, and suitability for particular applications. The comparison does not include body diodes of active switches or parasitic elements, but it does consider externally added components, such as input LC filters and snubber circuits, because they play significant roles in practical converter operation.

Selected zeta-derived and zeta-based DC-AC inverter studies are compared in Table 5. Most of the reported inverters are transformerless configurations, but a few studies use a single transformer to achieve galvanic isolation [139], [140]. Some diode-free designs are also considered [141], [142], [143], [144], [145], [146], [147], relying on active switching devices for power conversion without separate rectifier diodes. Both CCM and DCM are discussed in the surveyed literature, and some studies evaluate converter performance under either operating mode based on load and switching requirements. The reported power ratings range from 220 W [139] to 1 kW [141], [147], with several designs operating within the 432 W to 500 W range. Most systems maintain THD under 4%, with optimized designs reporting values near or below 1.1% [144] and 1% [148]. Reported efficiency ranges from 88% [148] to 97.58% [138]. The most common application areas are grid-connected PV systems, single-phase induction motor drives, BLDC motor drives, DC microgrids, and UPS systems.

Table 5. Evaluation of recent zeta-derived/zeta-based DC-AC inverter studies [D: diode; SW: switch; L: inductor; C: capacitor; TR: transformer]

Year	**D**	**SW**	**L**	**TR**	**C**	**Mode**	**Rated Power**	**THD**	**Efficiency**	**Applications**	**Study Focus**	**Ref.**
2015	4	4	4	0	4	CCM	Not Reported	8.95% @2.87 gain	Not Reported	General purpose	Design and simulation of a bidirectional differential zeta rectifier-inverter with buck-boost capability	[149]
2018	1	5	1	1	3	DCM and CCM	220 W	Not Reported	Not Reported	single-phase induction motor (SPIM)	Implementation of Fuzzy logic-based control for both MPPT of PV and	[139]

Year	D	SW	L	TR	C	Mode	Rated Power	THD	Efficiency	Applications	Study Focus	Ref.
											speed/torque of SPIM	
2019	4	4	3	0	4	DCM	1000 VA	2.2%	Not Reported	Grid-connected PV system	Design and analysis of Single-stage zeta inverter with hysteresis control and P&O MPPT	[150]
2020	0	7	5	0	1	Not Reported	1 kW	Not Reported	Not Reported	General purpose	Analysis of Soft-switching single-stage zeta-based inverter and SEPIC-based rectifier with small film capacitor	[141]
2021	0	5	2	1	3	DCM and CCM	300 W	< 1.7% (at full load)	95.76%	Distributed energy systems	Dynamic modelling and control of bridgeless hybrid mode zeta inverter.	[140]
2022	0	4	4	0	4	CCM	Not reported	Not reported	Not reported	General purpose	Analysis and comparison of Continuous and Discontinuous mode modulation scheme in differential mode zeta Inverter	[142]
2022	4	4	3	0	4	DCM	432 W	2.9 % @rate power	Not reported	Single phase grid tied PV systems	Design, modeling (static and dynamic) and small-signal analysis of modified zeta inverter	[41]
2022	0	7	4	0	1	Not Reported	600 W	Not reported	91-92%	BLDC motor, EV	Design and control of a bidirectional single-stage zeta-SEPIC with regenerative braking capability	[143]
2022	3	6	2	0	3	DCM	1 kW	3.4% (step-down condition) ; 3.9%	96.5%	PV system	Design and analysis of a single-stage common-ground zeta-based inverter	[42]

Year	D	SW	L	TR	C	Mode	Rated Power	THD	Efficiency	Applications	Study Focus	Ref.
								(step-up condition)			with nonelectrolytic capacitor	
2023	4	4	3	0	5	DCM	432 W	3.3%	Not Reported	Grid tied PV system	State-space modelling and multi loop controller design of SPIZ converter	[136]
2024	2	4	2	0	2	Not Reported	500W	3.48%	97.58 @300W/200V input	Grid connected system.	Design and validation of common grounded zeta-CSC inverter with leakage current elimination with bimodal controller design	[138]
2024	0	6	2	0	2	Not Reported	Not Reported	1.1%	Not Reported	renewable energy application	Design and validation of zeta H-bridge inverter using STM32F407 VET6	[144]
2024	0	4	2	0	2	Not Reported	Not Reported	PI : 1.81%; dq: 2.21%	Not Reported	highly nonlinear inductive loads	Low THD current control for nonlinear loads using dual ZETA inverter with GaN switches, comparing PI and dq-frame control strategies	[145]
2024	4	4	3	0	5	CCM	324 W	3.2%	95.06%	General purpose	Design and validation of SP-IZI with P-MR controller i	[137]
2025	4	4	3	0	3	CCM	450 VA	2.15% @ 1000 W/m²	94%	Grid tied PV system	Design and small signal modelling of IMZI with LQR control	[151]
2025	0	6	2	0	4	DCM	280 W	<3% (R load)	~90%	DC microgrids, UPS	Design and validation of IZCI using PI-MR control	[146]
2025	0	7	2	0	1	DCM	1 kW	2.8%	91%	low and medium-	Design, reliability analysis and	[147]

Year	D	SW	L	TR	C	Mode	Rated Power	THD	Efficiency	Applications	Study Focus	Ref.
										voltage systems	validation of a single-stage zeta-based AC-link universal converter.	

4.4. Zeta-Derived and Zeta-Based AC-AC Converters

Zeta-derived AC-AC converters extend the DC-DC Zeta energy-transfer principle to AC operation by replacing unidirectional switching elements with bidirectional switches and/or by using an AC-link implementation. Direct Zeta-mode AC-AC converters should be distinguished from Zeta-based AC-link universal converters, which can operate in AC-AC mode but are not exclusively AC-AC converters. Unlike conventional two-stage AC-DC-AC converters, which require bulky DC-link electrolytic capacitors, several Zeta-based AC-AC or AC-link converters use a small high-frequency film capacitor and magnetic energy-storage elements, improving reliability and extending converter lifetime [152], [153]. The topology family may employ bidirectional switches to allow current flow in both directions and can be implemented in single-phase or three-phase forms, with or without high-frequency transformer isolation. The converter can be designed to operate in CCM, achieving low THD and fast dynamic response, which makes it suitable for a wide range of power conversion applications [154].

Table 6 presents the performance and structural evaluation of selected zeta-derived and zeta-based AC-AC converter studies based on parameters such as power level, THD, power factor, efficiency, and component counts. Among the reviewed studies, the HID lamp electronic ballast in [155] includes an ignition transformer; this should not be interpreted as a power-stage isolation transformer unless galvanic isolation is explicitly claimed by the source. Most other topologies listed are transformerless. Some designs employ relatively high numbers of switches and diodes, including up to twelve of each in three-phase AC-link implementations [153], [156], [161], [152]. To ensure stable converter operation and good output performance, CCM operation is used in several reported works [35], [52], [154], [155], [157], [159].

The reported power ratings range from 70 W [155] to 1.2 kW [153], while multiple converters are designed around the 500 VA power level [35], [52], [157], [158], [159]. In terms of power quality, several studies report power factor values close to unity. The reported THD varies from 27.2% [155] to as low as 1% [152], depending on converter configuration and control methodology. The highest reported efficiency is 89.3% [155]. The examined AC-AC zeta-derived converters are primarily used in general-purpose power conversion systems, electronic ballasts for HID lamps [155], and automatic voltage regulation applications [160].

Table 6. Evaluation of recent zeta-derived/zeta-based AC-AC converter studies [D: diode; SW: switch; L: inductor; C: capacitor; TR: transformer]

Year	D	SW	L	TR	C	Mode	Rated Power	Power Factor	THD	Efficiency	Applications	Study Focus	Ref.
2009	10	10	3	0	3	Not Reported	500 VA	Not Reported	Not Reported	Not Reported	General purpose	Design and validation of zeta mode 3 level	[158]

Year	D	SW	L	TR	C	Mode	Rated Power	Power Factor	THD	Efficiency	Applications	Study Focus	Ref.
												AC/AC converter with non-complementary control strategy without dead-time effects	
2009	10	10	3	0	3	CCM	500 VA	Not Reported	Not Reported	Not Reported	high input voltage and power application	Design and validation of ZETA mode 3 level AC/AC converter with non complementary control	[35]
2010	8	2	2	1 (ignition transformer)	4	CCM	70 W	0.94	27.2%.	89.3 %	Electronic ballast for HID lamps	Design and validation of ZETA AC-AC converter to prevent acoustic resonance	[155]
2012	10	10	3	0	3	CCM	500 VA	Not Reported	Not Reported	Not Reported	General purpose	Output spectrum analysis and validation of zeta-mode three-level AC/AC converter	[159]
2012	10	10	3	0	3	CCM	500 VA	0.65 - 0.96 (normalized different nature loads)	< 3%	80.4% - 85.3% (normalized different nature loads)	General purpose	Design and validation of zeta-mode three-level AC direct converter with noncomplementary control	[52]
2022	12	12	3	0	4	Not Reported	1.2 kW	1	3.5%: output voltage; 4.8% input current	Not Reported	General purpose	Design of zeta based three phase AC/AC converter with small film capacitor	[153]
2022	4	2	3	0	3	Not Reported	Not Reported	Not Reported	Not Reported	Not Reported	Automatic AC voltage regulation	Design of single phase AC/AC zeta converter as AVR with delta-sigma modulation.	[160]
2023	12	12	3	0	5	Not Reported	1 kW	Not Reported	3.6%: output voltage; 4.2% input current	Not Reported	General purpose	Design and analysis of zeta based AC/AC converter with ZVS and ZCS	[156]
2023	12	12	3	0	3	Not Reported	1 kW	~1	2%: output voltage; 2.5% input current	Not Reported	General purpose	Design and validation of soft switching based AC/AC converter with elimination of electrolyte capacitors	[161]
2025	6	12	3	0	2	Not Reported	1 kW	~1	1%: output	Not Reported	low and medium-	Design, reliability analysis and	[152]

Year	D	SW	L	TR	C	Mode	Rated Power	Power Factor	THD	Efficiency	Applications	Study Focus	Ref.
									voltage; 1.7% input current		voltage systems	validation of a single-stage zeta-based AC-link universal converter.	

5. Future Potential and Challenges

The zeta has gained rapid popularity over the years as indicated by the increased academic publication illustrated in Fig 12.

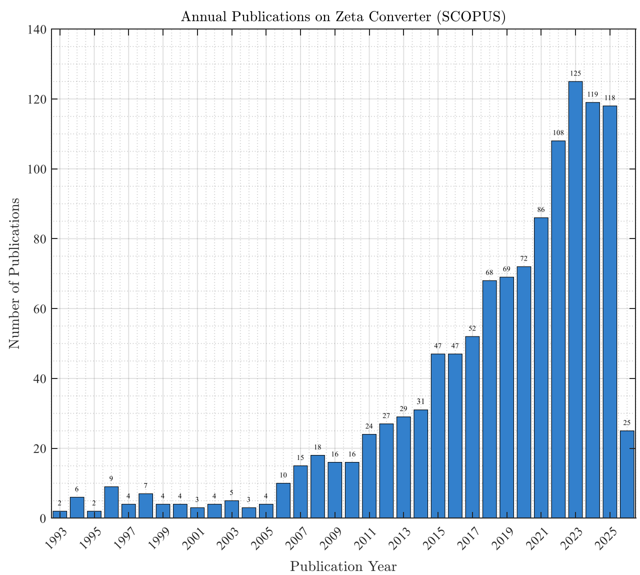


Fig 12: Publications on zeta converter over the year in SCOPUS [search criteria- keywords: "zeta" AND "converter" search area: title, abstract and keywords, search date: 22 May 2026]

Zeta converter topologies have considerable potential for adoption in a wide range of emerging applications. The major advantages and future research directions of Zeta converters are outlined below.

Renewable energy and grid applications: The inherent buck–boost capability and non-inverting output voltage make zeta converters well suited for PV, fuel cell, and hybrid renewable energy systems, where wide input voltage variations require efficient power conditioning. This is also illustrated in Fig. 13, which shows that one of the most common applications of the zeta converter is in standalone and grid-connected solar energy conversion systems. In addition, integrated zeta–SEPIC converter topologies can be designed to support bidirectional power flow, making them suitable for use in microgrids and energy storage applications.

Electric vehicles and battery energy storage systems: The ability to regulate voltage under fluctuating battery conditions makes zeta converters promising for battery management systems, auxiliary power supplies, on-board chargers, and energy storage interfaces. This is the most popular application where zeta and/or zeta derived topologies are currently being employed as displayed in Fig 13.

Wide-bandgap semiconductor integration: The adoption of wide-bandgap (WBG) semiconductor devices, such as gallium nitride (GaN) and silicon carbide (SiC), is expected to enable higher switching frequencies, reduced passive component size, improved power density, and higher conversion efficiency. These advantages are particularly beneficial for two-inductor DC–DC converter topologies, such as the zeta converter, as they enable significant reductions in converter size and weight while maintaining high efficiency. Furthermore, the higher switching frequencies achievable with WBG devices facilitate increased power density and improved thermal performance, making Zeta converters more attractive for compact renewable energy and electric mobility applications.

Emerging electrification applications: The growing adoption of electric transportation, portable power systems, aerospace electronics, and IoT devices presents new opportunities for zeta converters owing to their versatility, stable output characteristics, and ability to accommodate wide input voltage ranges.

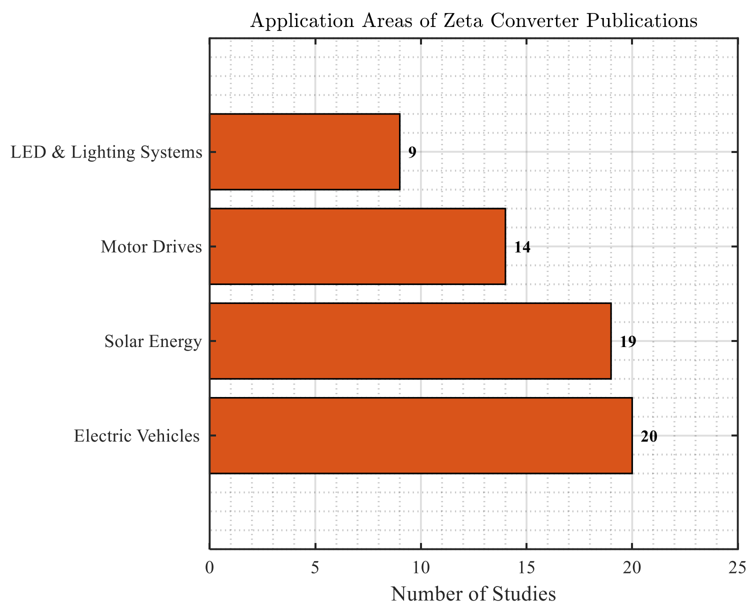


Fig 13: Distribution of the reviewed studies across different zeta converter application areas following the literature screening process

Despite their significant potential in energy conversion systems, zeta converters present several design challenges for researchers and engineers. The main issues and challenges associated with zeta converters are listed below:

Component count and converter complexity: The zeta converter requires two inductors and a coupling capacitor, resulting in a higher component count and increased design complexity compared with conventional buck and boost converters. This increases converter size, cost, and manufacturing complexity, particularly in high-power applications.

Efficiency and power losses: The presence of multiple passive components and higher circulating currents can increase conduction and switching losses, reducing overall efficiency under certain operating conditions. Further optimization of converter topology, magnetic design, and switching techniques is required to improve performance.

Control and dynamic performance: The fourth-order nature of the zeta converter leads to more complex control design and dynamic behavior than simpler DC–DC converters. Advanced digital control strategies are often required to ensure stable operation, fast transient response, and robust performance under varying load and input conditions.

High-power operation and scalability: Although the zeta converter is well suited for low- and medium-power applications, its implementation in high-power systems remains challenging because of increased component stresses, thermal management requirements, and EMI. Further research is needed to improve scalability while maintaining high efficiency and reliability.

Current and voltage stress: In bidirectional zeta converters, high current stress during DCM and significant voltage stress on the coupling capacitor can increase semiconductor device stress, potentially reducing efficiency and complicating high-power converter design

6. Conclusion

This survey paper presents a comprehensive overview of the topological evolution of the zeta converter, together with a systematic classification and comparative analysis of representative converter topologies based on their circuit configuration, component count, power rating, applications, and key performance metrics. By consolidating information that is otherwise scattered across the literature, the paper provides a structured reference that can assist researchers in identifying suitable zeta converter topologies for specific applications, understanding their respective advantages and limitations, and recognizing current research trends and development directions. It is also expected to facilitate future topology selection, comparative studies, and the development of improved zeta-based converter designs.

References

[1] T. Sutikno, R. A. Aprilianto, and H. S. Purnama, "Application of non-isolated bidirectional DC–DC converters for renewable and sustainable energy systems: a review," *Clean Energy*, vol. 7, no. 2, pp. 293–311, Apr. 2023, doi: 10.1093/ce/zkac070.

[2] F. L. Tofoli, T. M. J. Carlos, and A. S. Morais, "Review, Properties, and Synthesis of Single-Switch Non-Isolated DC-DC Converters with a Wide Conversion Range," *Sensors*, vol. 24, no. 7, p. 2264, Jan. 2024, doi: 10.3390/s24072264.

[3] L. R. Barbosa *et al.*, "Analysis of a Soft-single-switched Quadratic boost Converter," *Eletrônica Potência*, vol. 18, no. 3, pp. 1047–1054, Aug. 2013, doi: 10.18618/REP.2013.3.10471054.

[4] M. K. Kazimierczuk and J. J. Jozwik, "Optimal topologies of resonant DC/DC converters," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 25, no. 3, pp. 363–372, May 1989, doi: 10.1109/7.30791.

[5] F. Mumtaz, N. Zaihar Yahaya, S. Tanzim Meraj, B. Singh, R. Kannan, and O. Ibrahim, "Review on non-isolated DC-DC converters and their control techniques for renewable energy applications," *Ain Shams Eng. J.*, vol. 12, no. 4, pp. 3747–3763, Dec. 2021, doi: 10.1016/j.asej.2021.03.022.

[6] S. Rodríguez-Romero *et al.*, "Boost DC–DC Converter Topologies for ViPV Systems: A Technical Review," *IEEE Access*, vol. 14, pp. 56542–56571, 2026, doi: 10.1109/ACCESS.2026.3679258.

[7] S. M. Salehi and A. Y. Varjani, "A step-up DC-DC converter with high voltage gain and soft switched capability and minimum phase characteristic," *Sci. Rep.*, vol. 16, no. 1, p. 9763, Feb. 2026, doi: 10.1038/s41598-026-40326-8.

[8] I. Jagadeesh and V. Indragandhi, "Solar photo voltaic based hybrid CUK, SEPIC, ZETA converters for microgrid applications," *E-Prime - Adv. Electr. Eng. Electron. Energy*, vol. 6, p. 100364, Dec. 2023, doi: 10.1016/j.prime.2023.100364.

[9] D. Nayanasiri and Y. Li, "Step-Down DC–DC Converters: An Overview and Outlook." *Electronics*, vol. 11, no. 11, Art. no. 1693, 2022, doi: 10.3390/electronics11111693.

[10] N. Bagheri, H. Alipour, L. Mohammadian, J. Beiza, and M. Ebadpour, "A new high step-down soft-switched DC–DC converter based on coupled inductor: *Int. J. Electron.*, vol. 112, no. 7, pp. 1396–1416, 2025, doi: 10.1080/00207217.2024.2382493.

[11] M. R. S. de Carvalho, R. C. Neto, E. J. Barbosa, L. R. Limongi, F. Bradaschia, and M. C. Cavalcanti, "An Overview of Voltage Boosting Techniques and Step-Up DC-DC Converters Topologies for PV Applications" *Energies*, vol. 14, no. 24, Art. no. 8230, 2021, doi: 10.3390/en14248230.

[12] M. Forouzesh, Y. P. Siwakoti, S. A. Gorji, F. Blaabjerg and B. Lehman, “Step-Up DC–DC Converters: A Comprehensive Review of Voltage-Boosting Techniques, Topologies, and Applications.” *IEEE Transactions on Power Electronics*, vol. 32, no. 12, pp. 9143-9178, Dec. 2017, doi: 10.1109/TPEL.2017.2652318.

[13] D. V. S. Reddy and T. S, “Review on power electronic boost converters,” *Aust. J. Electr. Electron. Eng.*, vol. 18, no. 3, pp. 127–137, Jul. 2021, doi: 10.1080/1448837X.2021.1935091.

[14] H. Tarzamni, H. S. Gohari, M. Sabahi, and J. Kyyrä, “Nonisolated High Step-Up DC–DC Converters: Comparative Review and Metrics Applicability,” *IEEE Trans. Power Electron.*, vol. 39, no. 1, pp. 582–625, Jan. 2024, doi: 10.1109/TPEL.2023.3264172.

[15] F. L. Tofoli, D. D. C. Pereira, W. Josias De Paula, and D. D. S. Oliveira Júnior, “Survey on non-isolated high-voltage step-up dc–dc topologies based on the boost converter,” *IET Power Electron.*, vol. 8, no. 10, pp. 2044–2057, Oct. 2015, doi: 10.1049/iet-pel.2014.0605.

[16] P. Andrade, A. N. Alcaso, F. Bento, and A. J. Marques Cardoso, “Buck-Boost DC-DC Converters for Fuel Cell Applications in DC Microgrids—State-of-the-Art,” *Electronics*, vol. 11, no. 23, p. 3941, Nov. 2022, doi: 10.3390/electronics11233941.

[17] I. Ahmed, A. I. Alvy, I. Rahman, M. A. Al Hossain, S. Rahman, A. Al-Hysam, and A. Razzak, “AC-DC Cuk Converter: A State-of-the-Art topological review,” *Energy Convers. Manag. X*, vol. 27, doi: 10.1016/j.ecmx.2025.101115.

[18] T. M. K. Faistel, R. A. Guisso, A. M. S. S. Andrade, and M. L. Da Silva Martins, “Comparative evaluation of a family of isolated Ćuk DC/DC converter with step-up techniques,” *IET Power Electron.*, vol. 13, no. 16, pp. 3637–3650, Dec. 2020, doi: 10.1049/iet-pel.2020.0151.

[19] I. Ahmed, M. A. Niloy, M. Al Haque, S. Rahman, F. S. Azad, M. S. Hossain Lipu, A. M. T. Oo, and M. A. Razzak, “Recent advancement of AC–DC SEPIC converter: A state–of–the–art review on topologies, technical aspects and applications,” *Results Eng.*, vol. 27, p. 106814, Sep. 2025, doi: 10.1016/j.rineng.2025.106814.

[20] U. V V, S. George, and P. J. Paul, “A Comprehensive study on Traditional and Derived SEPIC Topologies,” in *2022 Third International Conference on Intelligent Computing Instrumentation and Control Technologies (ICICICT)*, Kannur, India: IEEE, Aug. 2022, pp. 1471–1476. doi: 10.1109/ICICICT54557.2022.9917661.

[21] H. A. Mosallam and E. A. Badran, “A systematic review of DC–DC converter technologies: traditional, bidirectional, and impedance-source topologies,” *Discov. Electron.*, vol. 3, no. 1, p. 16, Feb. 2026, doi: 10.1007/s44291-026-00168-0.

[22] S. A. Gorji, H. G. Sahebi, M. Ektesabi, and A. B. Rad, “Topologies and Control Schemes of Bidirectional DC–DC Power Converters: An Overview,” *IEEE Access*, vol. 7, pp. 117997–118019, 2019, doi: 10.1109/ACCESS.2019.2937239.

[23] I. Soltani, G. R. Molaeimanesh, H. Gholizadeh, and H. Vahedi, “A Review of High-Step-Up Non-Isolated DC–DC Converters Focusing on the Topology Methodology and Features,” *IEEE Access*, vol. 14, pp. 54320–54352, 2026, doi: 10.1109/ACCESS.2026.3681231.

[24] M. Vishvanath and A. Venkadesan, “A comprehensive review and comparative analysis of bridgeless power factor correction converter topologies for electric vehicle charging applications,” *Frankl. Open*, vol. 16, p. 100599, Sep. 2026, doi: 10.1016/j.fraope.2026.100599.

[25] Y. Tong, I. Salhi, Q. Wang, G. Lu, and S. Wu, “Bidirectional DC-DC Converter Topologies for Hybrid Energy Storage Systems in Electric Vehicles: A Comprehensive Review,” *Energies*, vol. 18, no. 9, p. 2312, May 2025, doi: 10.3390/en18092312.

[26] J. J. Jozwik and M. K. Kazimierczuk, “Dual sepic PWM switching-mode DC/DC power converter,” *IEEE Trans. Ind. Electron.*, vol. 36, no. 1, pp. 64–70, Feb. 1989, doi: 10.1109/41.20346.

[27] L. Hsiu, W. Kerwin, A. F. Witulski, R. Carlsten, and R. Ghotbi, “A coupled-inductor, zero-voltage-switched dual-SEPIC converter with low output ripple and noise,” in *[Proceedings] Fourteenth International Telecommunications Energy Conference - INTELEC ’92*, Washington, DC, USA: IEEE, 1992, pp. 186–193. doi: 10.1109/INTLEC.1992.268443.

[28] L. Hsiu, W. Kerwin, and A. F. Witulski, “Small signal models of a zero-voltage-switched SEPIC dual converter with a coupled inductor,” in *Proceedings of 1994 Power Electronics Specialist Conference - PESC’94*, Taipei, Taiwan: IEEE, 1994, pp. 814–820. doi: 10.1109/PESC.1994.373772.

[29] A. Peres, D. C. Martins, and I. Barbi, “ZETA converter applied in power factor correction,” in *Proceedings of 1994 Power Electronics Specialist Conference - PESC’94*, Taipei, Taiwan: IEEE, 1994, pp. 1152–1157. doi: 10.1109/PESC.1994.373827.

[30] Sam Ben-Yaakov, "The Zeta converter made simple and how did it evolve, Dec 4, 2016. Accessed: May 01, 2024. [Online]. Available: https://youtu.be/WPxQylJGJMA?si=PCEIu88zeicMjP1X - Google Search.” Accessed: Jun. 07, 2026. [Online].

[31] D. C. Martins and G. N. De Abreu, “Application of the ZETA converter in switched-mode power supplies,” in *Conference Record of the Power Conversion Conference - Yokohama 1993*, Yokohama, Japan: IEEE, 1993, pp. 147–152. doi: 10.1109/PCCON.1993.264232.

[32] M. K. Kazimierczuk and J. J. Jozwik, “Optimal topologies of resonant DC/DC converters,” *IEEE Trans. Aerosp. Electron. Syst.*, vol. 25, no. 3, pp. 363–372, May 1989, doi: 10.1109/7.30791.

[33] I.-D. Kim, S.-H. Paeng, J.-W. Ahn, E.-C. Nho, and J.-S. Ko, “New Bidirectional ZVS PWM Sepic/Zeta DC-DC Converter,” in *2007 IEEE International Symposium on Industrial Electronics*, Vigo, Spain: IEEE, Jun. 2007, pp. 555–560. doi: 10.1109/ISIE.2007.4374656.

[34] Bor-Ren Lin and Fang-Yu Hsieh, “Soft-Switching Zeta–Flyback Converter With a Buck–Boost Type of Active Clamp,” *IEEE Trans. Ind. Electron.*, vol. 54, no. 5, pp. 2813–2822, Oct. 2007, doi: 10.1109/TIE.2007.901366.

[35] Q. Zhong and L. Li, “Zeta Mode Three-level AC/AC Converter,” in *2009 Twenty-Fourth Annual IEEE Applied Power Electronics Conference and Exposition*, Washington, DC, USA: IEEE, Feb. 2009, pp. 1831–1834. doi: 10.1109/APEC.2009.4802919.

[36] R. C. Viero, H. F. M. Lopez, C. A. Zollmann, and F. S. Dos Reis, “Dynamic modeling of a sinusoidal inverter based on ZETA converter working in DCM for PV arrays,” in *IECON 2010 - 36th Annual Conference on IEEE Industrial Electronics Society*, Glendale, AZ, USA: IEEE, Nov. 2010, pp. 439–444. doi: 10.1109/IECON.2010.5674986.

[37] B. Singh and V. Bist, “A PFC based BLDC motor drive using a Bridgeless Zeta converter,” in *IECON 2013 - 39th Annual Conference of the IEEE Industrial Electronics Society*, Vienna, Austria: IEEE, Nov. 2013, pp. 2553–2558. doi: 10.1109/IECON.2013.6699533.

[38] V. Bist and B. Singh, “A reduced sensor PFC BL-Zeta converter based VSI fed BLDC motor drive,” *Electr. Power Syst. Res.*, vol. 98, pp. 11–18, May 2013, doi: 10.1016/j.epsr.2013.01.006.

[39] V. Bist and B. Singh, “A Brushless DC Motor Drive With Power Factor Correction Using Isolated Zeta Converter,” *IEEE Trans. Ind. Inform.*, vol. 10, no. 4, pp. 2064–2072, Nov. 2014, doi: 10.1109/TII.2014.2346689.

[40] R. K. Surapaneni and A. K. Rathore, “A Single-Stage CCM Zeta Microinverter for Solar Photovoltaic AC Module,” *IEEE J. Emerg. Sel. Top. Power Electron.*, vol. 3, no. 4, pp. 892–900, Dec. 2015, doi: 10.1109/JESTPE.2015.2438012.

[41] L. P. Sampaio, P. J. S. Costa, and S. A. O. Da Silva, “Modified zeta inverter intended for single-phase grid-tied photovoltaic system,” *Sustain. Energy Technol. Assess.*, vol. 52, p. 102076, Aug. 2022, doi: 10.1016/j.seta.2022.102076.

[42] L. Wang and M. Shan, “A Novel Single-Stage Common-Ground Zeta-Based Inverter With Nonelectrolytic Capacitor,” *IEEE Trans. Power Electron.*, vol. 37, no. 9, pp. 11319–11331, Sep. 2022, doi: 10.1109/TPEL.2022.3167450.

[43] D. C. Martins and M. M. Casaro, “Isolated three-phase rectifier with high power factor using the Zeta converter in continuous conduction mode,” *IEEE Trans. Circuits Syst. Fundam. Theory Appl.*, vol. 48, no. 1, pp. 74–80, Jan. 2001, doi: 10.1109/81.903189.

[44] J.-L. Lin, S.-P. Yang, and P.-W. Lin, “Small-signal analysis and controller design for an isolated zeta converter with high power factor correction,” *Electr. Power Syst. Res.*, vol. 76, no. 1–3, pp. 67–76, Sep. 2005, doi: 10.1016/j.epsr.2005.04.008.

[45] B. Singh, B. P. Singh, and S. Dwivedi, “AC-DC Zeta converter for power quality improvement in direct torque controlled PMSM drive.” *Journal of Power Electronics*, vol. 6, no. 2, pp. 146–162, Apr. 2006.

[46] S. Narula, B. Singh, and G. Bhuvaneswari, “Power quality improved SMPS using bridgeless isolated zeta converter for welding applications,” *EPE J.*, vol. 26, no. 4, pp. 153–164, Oct. 2016, doi: 10.1080/09398368.2016.1272211.

[47] A. A. Vikram, R. Navaneeth, M. N. Kumar, and R. Vinoth, “Solar PV Array Fed BLDC Motor Using Zeta Converter For Water Pumping Applications,” *J. Sci. Technol.*, vol. 3, no. 4, pp. 08–19, Jul. 2018.

[48] H. F. Lopez, C. Zollmann, and R. V. E. F. S. D. Reis, “Photovoltaic panels grid-tied by a Zeta converter,” in *2009 Brazilian Power Electronics Conference*, Bonito-Mato Grosso do Sul, Brazil: IEEE, Sep. 2009, pp. 181–188. doi: 10.1109/COBEP.2009.5347658.

[49] A. K. Mishra and B. Singh, “SPV array powered zeta converter fed SRM drive for water pumping,” in *2015 IEEE Power, Communication and Information Technology Conference (PCITC)*, Bhubaneswar, India: IEEE, Oct. 2015, pp. 476–482. doi: 10.1109/PCITC.2015.7438213.

[50] B. Saha and B. Singh, “Sensorless PMBLDC Motor Driven Solar-PV Battery Fed EV System With Regenerative Braking Employing Zeta Converter,” in *2021 International Conference on Sustainable Energy and Future Electric Transportation (SEFET)*, Hyderabad, India: IEEE, Jan. 2021, pp. 1–6. doi: 10.1109/SeFet48154.2021.9375728.

[51] N. Jeelani and A. Hamid Bhat, “Constant switching frequency-based delta-sigma modulation of single-phase AC–AC zeta converter,” *Int. J. Circuit Theory Appl.*, vol. 51, no. 1, pp. 97–114, Jan. 2023, doi: 10.1002/cta.3435.

[52] L. Li and Q. Zhong, “Novel Zeta-Mode Three-Level AC Direct Converter,” *IEEE Trans. Ind. Electron.*, vol. 59, no. 2, pp. 897–903, Feb. 2012, doi: 10.1109/TIE.2011.2161655.

[53] M. J. Bonato, F. T. Wakabayashi, and C. A. Canesin, “A novel voltage step-down/up ZCS-PWM Zeta converter,” in *Conference Record of the 2000 IEEE Industry Applications Conference. Thirty-Fifth IAS Annual Meeting and World Conference on Industrial Applications of Electrical Energy (Cat. No.00CH37129)*, Rome, Italy: IEEE, 2000, pp. 2448–2454. doi: 10.1109/IAS.2000.883166.

[54] S. H. Hosseini, R. S. Alishah, and N. V. Kurdkandi, “Design of a new extended zeta converter with high voltage gain for photovoltaic applications,” in *2015 9th International Conference on Power Electronics and ECCE Asia (ICPE-ECCE Asia)*, Seoul, South Korea: IEEE, Jun. 2015, pp. 970–977. doi: 10.1109/ICPE.2015.7167899.

[55] E. Oh, M. S. Song, B. Kang, and Y. K. Park, “P-59: High-Efficiency Direct-Linked Zeta Converter for LED Backlight Units,” *SID Symp. Dig. Tech. Pap.*, vol. 42, no. 1, pp. 1320–1323, Jun. 2011, doi: 10.1889/1.3621081.

[56] J. Daniel Sathyaraj, R. Arumugam, and M. Faustino Adlinde, “A novel interleaved Zeta–Cuk converter for microgrid and electric vehicle applications,” *Electr. Eng.*, vol. 105, no. 6, pp. 4177–4193, Dec. 2023, doi: 10.1007/s00202-023-01951-y.

[57] V. Chapparya, A. Dey, and S. P. Singh, “A Novel Non-Isolated Boost-Zeta Interleaved DC-DC Converter for Low Voltage Bipolar DC Micro-Grid Application,” *IEEE Trans. Ind. Appl.*, vol. 59, no. 5, pp. 6182–6192, Sep. 2023, doi: 10.1109/TIA.2023.3284810.

[58] A. M. S. S. Andrade and M. L. D. S. Martins, “Quadratic-Boost With Stacked Zeta Converter for High Voltage Gain Applications,” *IEEE J. Emerg. Sel. Top. Power Electron.*, vol. 5, no. 4, pp. 1787–1796, Dec. 2017, doi: 10.1109/JESTPE.2017.2706220.

[59] B. Zhu, G. Liu, Y. Zhang, Y. Huang, and S. Hu, “Single-Switch High Step-Up Zeta Converter Based on Coat Circuit,” *IEEE Access*, vol. 9, pp. 5166–5176, 2021, doi: 10.1109/ACCESS.2020.3048388.

[60] M. S. Bhaskar *et al.*, “A New Hybrid Zeta-Boost Converter With Active Quad Switched Inductor for High Voltage Gain,” *IEEE Access*, vol. 9, pp. 20022–20034, 2021, doi: 10.1109/ACCESS.2021.3054393.

[61] P. L. Santosh Kumar Reddy, Y. P. Obulesu, S. Singirikonda, M. Al Harthi, M. S. Alzaidi, and S. S. M. Ghoneim, “A Non-Isolated Hybrid Zeta Converter with a High Voltage Gain and Reduced Size of Components,” *Electronics*, vol. 11, no. 3, p. 483, Feb. 2022, doi: 10.3390/electronics11030483.

[62] M.-S. Song, Y.-D. Son, and K.-H. Lee, “Non-isolated Bidirectional Soft-switching SEPIC/ZETA Converter with Reduced Ripple Currents,” *J. Power Electron.*, vol. 14, no. 4, pp. 649–660, Jul. 2014, doi: 10.6113/JPE.2014.14.4.649.

[63] V. Mahendran and R. Ramabadran, “Implementation of zero voltage switched SEPIC/ZETA bidirectional converter for low power applications using FPGA,” *Turk. J. Electr. Eng. Comput. Sci.*, vol. 25, pp. 319–336, 2017, doi: 10.3906/elk-1505-81.

[64] P. K. Gayen, P. Roy Chowdhury, and P. K. Dhara, “An improved dynamic performance of bidirectional SEPIC-Zeta converter based battery energy storage system using adaptive sliding mode control technique,” *Electr. Power Syst. Res.*, vol. 160, pp. 348–361, Jul. 2018, doi: 10.1016/j.epsr.2018.03.016.

[65] E. E. Henao-Bravo, A. J. Saavedra-Montes, C. A. Ramos-Paja, J. D. Bastidas-Rodriguez, and D. Gonzalez Montoya, “Charging/discharging system based on zeta/sepic converter and a sliding mode controller for dc bus voltage regulation,” *IET Power Electron.*, vol. 13, no. 8, pp. 1514–1527, Jun. 2020, doi: 10.1049/iet-pel.2019.0746.

[66] B. Xu *et al.*, “A Bidirectional Integrated Equalizer Based on the Sepic–Zeta Converter for Hybrid Energy Storage System,” *IEEE Trans. Power Electron.*, vol. 37, no. 10, pp. 12659–12668, Oct. 2022, doi: 10.1109/TPEL.2022.3161984.

[67] N. Vosoughi, M. Abbasi, E. Abbasi, and M. Sabahi, “A Zeta-based switched-capacitor DC-DC converter topology,” *Int. J. Circuit Theory Appl.*, vol. 47, no. 8, pp. 1302–1322, Aug. 2019, doi: 10.1002/cta.2647.

[68] M. R. Banaei and H. A. F. Bonab, “A High Efficiency Nonisolated Buck–Boost Converter Based on ZETA Converter,” *IEEE Trans. Ind. Electron.*, vol. 67, no. 3, pp. 1991–1998, Mar. 2020, doi: 10.1109/TIE.2019.2902785.

[69] S. Markkassery, A. Saradagi, A. D. Mahindrakar, L. N, and R. Pasumarthy, “Modelling, design and control of non-isolated single-input multi-output Zeta-Buck-Boost converter,” *IEEE Trans. Ind. Appl.*, pp. 1–1, 2020, doi: 10.1109/TIA.2020.2984190.

[70] D. Kumar, R. A. Gupta, and H. Tiwari, “FRONT-END zeta converter based BLDCM drive for efficient reduction of commutation current ripple using NOTCH-FILTER,” *Int. Trans. Electr. Energy Syst.*, vol. 30, no. 9, Sep. 2020, doi: 10.1002/2050-7038.12508.

[71] H. Ghojavand and E. Adib, “A Single-Switch Soft-Switched High Step-Down Zeta Converter With Low Output Current Ripple,” *IEEE Trans. Power Electron.*, vol. 38, no. 4, pp. 5024–5031, Apr. 2023, doi: 10.1109/TPEL.2022.3232388.

[72] X. Sun, D. Rong, and N. Wang, “A High Step-Up Integrated Buck-Boost-Zeta Converter Using Three-Winding Coupled Inductor With Current Sharing Considered,” *IEEE J. Emerg. Sel. Top. Power Electron.*, vol. 11, no. 3, pp. 3323–3334, Jun. 2023, doi: 10.1109/JESTPE.2023.3249345.

[73] S. R. Hole and A. D. Goswami, "Design of a novel hybrid soft computing model for passive components selection in multiple load Zeta converter topologies of solar PV energy system," *Energy Harvest. Syst.*, vol. 11, no. 1, p. 20230029, Jan. 2024, doi: 10.1515/ehs-2023-0029.

[74] P. M. S. Angeline and M. N. Rajkumar, "Integration of electric vehicle with smart grid using bidirectional SEPIC–zeta converter," *Electr. Eng.*, vol. 106, no. 2, pp. 2159–2174, Apr. 2024, doi: 10.1007/s00202-023-02116-7.

[75] Sugunakar Mamidala and Pavan Kumar Y. V., "Closed loop battery current controlled zeta converter for improved power quality in electric vehicle charging stations," *Trans. Energy Syst. Eng. Appl.*, vol. 5, no. 2, pp. 1–19, Oct. 2024, doi: 10.32397/tesea.vol5.n2.569.

[76] Y. He, L. Chen, and X. Sun, "An Interleaved Buck-Boost-Zeta Converter With Coupled Inductor Multiplier Cell and Zero Input Current Ripple for High Step-Up Applications," *IEEE Access*, vol. 12, pp. 104807–104817, 2024, doi: 10.1109/ACCESS.2024.3435140.

[77] M. F. Adlinde, A. Ravi, and J. D. Sathyaraj, "Enhanced power generation and management in hybrid PV-wind microgrid with modified Z-source Zeta converter and battery storage," *Electr. Eng.*, vol. 106, no. 4, pp. 5007–5026, Aug. 2024, doi: 10.1007/s00202-024-02263-5.

[78] K. Chen, B. Xu, and L. Liu, "A charging equalizer based on Zeta converter for hybrid energy storage system," *IET Power Electron.*, vol. 17, no. 3, pp. 463–472, Feb. 2024, doi: 10.1049/pel2.12661.

[79] Shandong University of Science and Technology and X. Fang, "The High Voltage Gain Coupled Inductor Boost-Zeta DC/DC Converter," *CPSS Trans. Power Electron. Appl.*, vol. 9, no. 2, Jun. 2024, doi: 10.24295/CPSSTPEA.2024.00001.

[80] M. Hosseinpour, M. Heydarvand, and M. E. Azizkandi, "A new positive output DC–DC buck–boost converter based on modified boost and ZETA converters," *Sci. Rep.*, vol. 14, no. 1, p. 20675, Sep. 2024, doi: 10.1038/s41598-024-71612-y.

[81] M. B. E. Kattel, R. Saraiva De Santiago, R. Mayer, E. Mineiro Sa, L. H. S. C. Barreto, and F. Luiz Marcelo Antunes, "DC–DC Multi-Arm Interleaved Zeta Converter With Parallel Switch Modulation for Renewable-Powered Electrolyzers," *IEEE Access*, vol. 13, pp. 166760–166777, 2025, doi: 10.1109/ACCESS.2025.3612331.

[82] M. J. Mohammed *et al.*, "A Control Strategy for Zeta Converter Adaptation in Photovoltaic Systems: Genetic Algorithm-Based PID Controller," *Int. J. Robot. Control Syst.*, vol. 5, no. 6, pp. 3376–3394, Jan. 2026, doi: 10.31763/ijrcs.v5i6.2079.

[83] D. Angulo-García, F. Angulo, and J.-G. Muñoz, "Robust Control of a Zeta Converter: A Feedback Linearization Approach with Digital PWM Implementation," *Energies*, vol. 18, no. 8, p. 1927, Apr. 2025, doi: 10.3390/en18081927.

[84] S. Mishra and S. Singh, "Performance Assessment of Bridgeless Zeta Converter and Solar Energized Zeta Converter Based ACS," *J. Sci. Res.*, vol. 17, no. 2, pp. 507–518, May 2025, doi: 10.3329/jsr.v17i2.77859.

[85] K. S. Rani and V. Nandagopal, "Comparative analysis of zeta converter and conventional DC-DC converter using mother optimized FOPID controller in electric vehicles," *Electr. Power Syst. Res.*, vol. 252, p. 112262, Jan. 2026, doi: 10.1016/j.epsr.2025.112262.

[86] J.-B. Lee, K.-B. Park, J.-K. Kim, H.-S. Youn, and G.-W. Moon, "A New Center-Tapped Half-Bridge Zeta Converter With Small Transformer DC-Offset Current and Low Voltage Stress," *IEEE Trans. Power Electron.*, vol. 30, no. 12, pp. 6593–6603, Dec. 2015, doi: 10.1109/TPEL.2015.2390254.

[87] J. Baek, C.-E. Kim, and J.-B. Lee, "Semiresonant Half-Bridge Zeta Converter With Reduced Turn-OFF Currents for High-Efficiency Power Supply," *IEEE Trans. Power Electron.*, vol. 35, no. 5, pp. 4547–4557, May 2020, doi: 10.1109/TPEL.2019.2943027.

[88] Bor-Ren Lin and Fang-Yu Hsieh, "Soft-Switching Zeta–Flyback Converter With a Buck–Boost Type of Active Clamp," *IEEE Trans. Ind. Electron.*, vol. 54, no. 5, pp. 2813–2822, Oct. 2007, doi: 10.1109/TIE.2007.901366.

[89] D. Murthy-Bellur and M. K. Kazimierczuk, "Isolated Two-Transistor Zeta Converter With Reduced Transistor Voltage Stress," *IEEE Trans. Circuits Syst. II Express Briefs*, vol. 58, no. 1, pp. 41–45, Jan. 2011, doi: 10.1109/TCSII.2010.2092829.

[90] Tsai-Fu Wu, Shih-An Liang, and Yaow-Ming Chen, "Design optimization for asymmetrical ZVS-PWM zeta converter," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 39, no. 2, pp. 521–532, Apr. 2003, doi: 10.1109/TAES.2003.1207264.

[91] B.-R. Lin and J.-J. Chen, "Analysis of an integrated flyback and zeta converter with active clamping technique," *IET Power Electron.*, vol. 2, no. 4, pp. 355–363, Jul. 2009, doi: 10.1049/iet-pel.2008.0038.

[92] A. Ruseler and I. Barbi, "Isolated Zeta-SEPIC bidirectional dc-dc converter with active-clamping," in *2013 Brazilian Power Electronics Conference*, Gramado, Brazil: IEEE, Oct. 2013, pp. 123–128. doi: 10.1109/COBEP.2013.6785104.

[93] S.-Y. Tseng and J.-H. Fan, "Zeta/Flyback Hybrid Converter for Solar Power Applications," *Sustainability*, vol. 14, no. 5, p. 2924, Mar. 2022, doi: 10.3390/su14052924.

[94] Ki-Bum Park, Chong-Eun Kim, Gun-Woo Moon, and Myung-Joong Youn, "A Double-Ended ZVS Half-Bridge Zeta Converter," *IEEE Trans. Power Electron.*, vol. 23, no. 6, pp. 2838–2846, Nov. 2008, doi: 10.1109/TPEL.2008.2003189.

[95] J.-B. Lee, K.-B. Park, H.-S. Kim, H.-W. Seong, G.-W. Moon, and M.-J. Youn, "ZVS half-bridge zeta converter with center-tapped rectifier," in *8th International Conference on Power Electronics - ECCE Asia*, Jeju, Korea (South): IEEE, May 2011, pp. 1606–1611. doi: 10.1109/ICPE.2011.5944383.

[96] H.-Y. Lee, T.-J. Liang, J.-F. Chen, and K.-H. Chen, "Design and implementation of a bidirectional SEPIC-Zeta DC-DC Converter," in *2014 IEEE International Symposium on Circuits and Systems (ISCAS)*, Melbourne VIC, Australia: IEEE, Jun. 2014, pp. 101–104. doi: 10.1109/ISCAS.2014.6865075.

[97] J.-I. Baek, J.-K. Kim, J.-B. Lee, H.-S. Youn, and G.-W. Moon, "Integrated Asymmetrical Half-Bridge Zeta (AHBZ) Converter for DC/DC Stage of LED Driver With Wide Output Voltage Range and Low Output Current," *IEEE Trans. Ind. Electron.*, vol. 62, no. 12, pp. 7489–7498, Dec. 2015, doi: 10.1109/TIE.2015.2454486.

[98] J.-I. Baek *et al.*, "PWM half-bridge zeta converter utilizing resonant technique for reduced peak current," in *2017 IEEE 3rd International Future Energy Electronics Conference and ECCE Asia (IFEEC 2017 - ECCE Asia)*, Kaohsiung, Taiwan: IEEE, Jun. 2017, pp. 483–487. doi: 10.1109/IFEEC.2017.7992086.

[99] J.-B. Lee, K.-B. Park, J.-K. Kim, H.-S. Youn, and G.-W. Moon, "A New Center-Tapped Half-Bridge Zeta Converter With Small Transformer DC-Offset Current and Low Voltage Stress," *IEEE Trans. Power Electron.*, vol. 30, no. 12, pp. 6593–6603, Dec. 2015, doi: 10.1109/TPEL.2015.2390254.

[100] B. S. Prajapati and R. K. Singh, "Modeling and Design of Triple Transistor Isolation-based Bidirectional Zeta-SEPIC DC-DC Converter," in *2022 IEEE International Conference on Power Electronics, Drives and Energy Systems (PEDES)*, Jaipur, India: IEEE, Dec. 2022, pp. 1–8. doi: 10.1109/PEDES56012.2022.10080353.

[101] B. S. Prajapati and R. K. Singh, "Modified hybrid isolated zeta–boost converter for electric vehicle charging from low-voltage renewable sources," *J. Power Electron.*, vol. 25, no. 4, pp. 619–630, Apr. 2025, doi: 10.1007/s43236-024-00927-w.

[102] H. Zhang, Y. Zhang, and X. Ma, "Distortion Behavior Analysis of General Pulse-Width Modulated Zeta PFC Converter Operating in Continuous Conduction Mode," *IEEE Trans. Power Electron.*, vol. 27, no. 10, pp. 4212–4223, Oct. 2012, doi: 10.1109/TPEL.2012.2191161.

[103] Ke-Ming Chen, Tsorng-Juu Liang, Shih-Ming Chen, and Kai-Hui Chen, "Design and implementation of a interleaved single-phase power factor correction Zeta converter," in *2013 IEEE 10th International Conference on Power Electronics and Drive Systems (PEDS)*, Kitakyushu: IEEE, Apr. 2013, pp. 171–174. doi: 10.1109/PEDS.2013.6527009.

[104] "(PDF) AC-DC Zeta converter for power quality improvement in direct torque controlled PMSM drive," *ResearchGate*, Accessed: Jun. 08, 2026. [Online]. Available: https://www.researchgate.net/publication/228624619_AC-DC_Zeta_converter_for_power_quality_improvement_in_direct_torque_controlled_PMSM_drive

[105] S. Pal, B. Singh, and A. Shrivastava, "Constant DC-bus critical conduction mode Zeta converter fed primary-side regulated LLC resonant converter in LED lighting applications," *Int. J. Circuit Theory Appl.*, vol. 49, no. 12, pp. 4142–4161, Dec. 2021, doi: 10.1002/cta.3096.

[106] S. S. Sharma, S. K. Sharma, V. K. Dubey, and R. Saxena, "Zeta Converter-Based Dimmable LED Light System," *J. Inst. Eng. India Ser. B*, vol. 104, no. 4, pp. 953–961, Aug. 2023, doi: 10.1007/s40031-023-00900-4.

[107] A. K. Singh and M. K. Pathak, "Single-stage ZETA-SEPIC-based multifunctional integrated converter for plug-in electric vehicles," *IET Electr. Syst. Transp.*, vol. 8, no. 2, pp. 101–111, Jun. 2018, doi: 10.1049/iet-est.2017.0063.

[108] A. Jha and B. Singh, "Bridgeless ZETA PFC converter for low voltage high current LED driver," in *2017 6th International Conference on Computer Applications In Electrical Engineering-Recent Advances (CERA)*, Roorkee, India: IEEE, Oct. 2017, pp. 539–544. doi: 10.1109/CERA.2017.8343387.

[109] V. Bist and B. Singh, "Reduced sensor configuration of brushless DC motor drive using a power factor correction-based modified-zeta converter," *IET Power Electron.*, vol. 7, no. 9, pp. 2322–2335, Sep. 2014, doi: 10.1049/iet-pel.2013.0177.

[110] A. M. El-Shebiny, H. A. Yasin, and S. S. Shokralla, "DYNAMIC ANALYSIS AND SPEED CONTROL OF AC-DC ZETA CONVERTER FED DC MOTOR," *ERJ Eng. Res. J.*, vol. 31, no. 1, pp. 1–6, Jan. 2008, doi: 10.21608/erjm.2008.69489.

[111] "(PDF) AC-DC Zeta converter for power quality improvement in direct torque controlled PMSM drive," *ResearchGate*, Accessed: Jun. 08, 2026. [Online]. Available: https://www.researchgate.net/publication/228624619_AC-DC_Zeta_converter_for_power_quality_improvement_in_direct_torque_controlled_PMSM_drive

[112] M. R. T. Hossain, A. H. Abedin, Md. H. Rahaman, Md. S. Arifin, M. A. Choudhury, and M. N. Uddin, “Input switched single phase high performance bridgeless AC-DC Zeta converter,” in *2012 IEEE International Conference on Power Electronics, Drives and Energy Systems (PEDES)*, Bengaluru, Karnataka, India: IEEE, Dec. 2012, pp. 1–5. doi: 10.1109/PEDES.2012.6484335.

[113] M. Shunmathi, J. Gnanavadivel, and S. T. Jaya Christa, “Power Factor Correction With Single Stage AC-DC Non-Isolated Zeta Converter Fed SRM Drive,” in *2021 7th International Conference on Electrical Energy Systems (ICEES)*, Chennai, India: IEEE, Feb. 2021, pp. 95–100. doi: 10.1109/ICEES51510.2021.9383693.

[114] A. D. Kumar, J. Gupta, and B. Singh, “A Modified Zeta HPF AC-DC Converter with High Step- Down Gain using Switched Capacitor and Coupled Inductor Structures for LVEVs Charging Applications,” in *2022 IEEE 10th Power India International Conference (PIICON)*, New Delhi, India: IEEE, Nov. 2022, pp. 1–6. doi: 10.1109/PIICON56320.2022.10045213.

[115] V. M, A. Venkadesan, and S. S. Sakthivel, “Design and analysis of bridgeless zeta-derived power factor correction converter with reduced magnetic components,” *Comput. Electr. Eng.*, vol. 123, p. 110017, Apr. 2025, doi: 10.1016/j.compeleceng.2024.110017.

[116] R. Thangasankaran, “Design of a High Efficiency AC-DC Converter with Interleaved Zeta Topology for Electric Vehicle off Board Chargers,” *Int. J. Res. Appl. Sci. Eng. Technol.*, vol. 13, no. 3, pp. 3462–3469, Mar. 2025, doi: 10.22214/ijraset.2025.68108.

[117] V. M, V. A, and J. Thomas, “Performance analysis of a multi-port bridgeless zeta converter topology,” *Eng. Res. Express*, vol. 7, no. 3, p. 035360, Sep. 2025, doi: 10.1088/2631-8695/adfe3c.

[118] R. Kushwaha, V. Khadkikar, M. Shawky El Moursi, and H. Zeineldin, “An On-Board Charger Using New Bridgeless PFC Converter With Reduced DC-Link Capacitance,” *IEEE Trans. Ind. Appl.*, vol. 61, no. 4, pp. 6695–6706, Jul. 2025, doi: 10.1109/TIA.2025.3543783.

[119] R. Kushwaha and B. Singh, “Bridgeless Isolated Zeta–Luo Converter-Based EV Charger With PF Preregulation,” *IEEE Trans. Ind. Appl.*, vol. 57, no. 1, pp. 628–636, Jan. 2021, doi: 10.1109/TIA.2020.3036019.

[120] B. Singh, M. Agrawal, and S. Dwivedi, “Analysis, Design, and Implementation of a Single-Phase Power-Factor Corrected AC-DC Zeta Converter with High Frequency Isolation,” *J. Electr. Eng. Technol.*, vol. 3, no. 2, pp. 243–253, Jun. 2008, doi: 10.5370/JEET.2008.3.2.243.

[121] A. D. Callegaro, D. C. Martins, and I. Barbi, “Isolated single-phase high power factor rectifier using Zeta converter operating in DCM with non-dissipative snubber,” in *2013 Brazilian Power Electronics Conference*, Gramado, Brazil: IEEE, Oct. 2013, pp. 1–6. doi: 10.1109/COBEP.2013.6785087.

[122] S. Narula, B. Singh, and G. Bhuvaneswari, “Power Factor Corrected Welding Power Supply Using Modified Zeta Converter,” *IEEE J. Emerg. Sel. Top. Power Electron.*, vol. 4, no. 2, pp. 617–625, Jun. 2016, doi: 10.1109/JESTPE.2015.2500610.

[123] A. Anand and B. Singh, “Isolated Zeta Converter Fed SRM Drive with High Power Conversion Efficiency,” in *2018 5th IEEE Uttar Pradesh Section International Conference on Electrical, Electronics and Computer Engineering (UPCON)*, Gorakhpur: IEEE, Nov. 2018, pp. 1–6. doi: 10.1109/UPCON.2018.8597083.

[124] A. Kumar, U. Sharma, and B. Singh, “PMBLDC Motor Based Ceiling Fan Using an Isolated PFC Zeta Converter,” in *2020 IEEE International Conference on Power Electronics, Smart Grid and Renewable Energy (PESGRE2020)*, Cochin, India: IEEE, Jan. 2020, pp. 1–6. doi: 10.1109/PESGRE45664.2020.9070675.

[125] “Fuzzy Logic Controller Based Bridgeless Isolated Interleaved Zeta Converter for LED Lamp Driver Application - ProQuest.” Accessed: Jun. 08, 2026. [Online]. Available: https://www.proquest.com/openview/86f0fd24de21479a6e19fd44a906e772/1?pq-origsite=gscholar&cbl=1686343

[126] B. Anil Kumar, S. Sasikanth, and M. Kiran Kumar, “An Isolated Zeta Converter Topology for Switched Reluctance Motor Drive with Power Factor Correction,” *Indian J. Sci. Technol.*, vol. 9, no. 33, Sep. 2016, doi: 10.17485/ijst/2016/v9i33/99490.

[127] B. Singh, B. P. Singh, and S. Dwivedi, “Performance Comparison of High Frequency Isolated AC-DC Converters for Power Quality Improvement at Input AC Mains,” in *2006 International Conference on Power Electronic, Drives and Energy Systems*, IEEE, Dec. 2006, pp. 1–6. doi: 10.1109/PEDES.2006.344401.

[128] P. Kochcha and S. Sujitjorn, “Isolated Zeta Converter: Principle of Operation and Design in Continuous Conduction Mode”.

[129] “Fuzzy Logic Controller Based Bridgeless Isolated Interleaved Zeta Converter for LED Lamp Driver Application - ProQuest.” Accessed: Jun. 08, 2026. [Online]. Available: https://www.proquest.com/openview/86f0fd24de21479a6e19fd44a906e772/1?pq-origsite=gscholar&cbl=1686343

[130] B. Singh and R. Kushwaha, “An EV battery charger with power factor corrected bridgeless zeta converter topology,” in *2016 7th India International Conference on Power Electronics (IICPE)*, Patiala, India: IEEE, Nov. 2016, pp. 1–6. doi: 10.1109/IICPE.2016.8079359.

[131] R. Narayan Deo, A. Shrivastava, and K. Chatterjee, "Single Stage Isolated Zeta Converter (SS-IZC) Circuit with Improved Power Factor for LED's Application," *IOP Conf. Ser. Mater. Sci. Eng.*, vol. 594, no. 1, p. 012014, Aug. 2019, doi: 10.1088/1757-899X/594/1/012014.

[132] S. T. Siddharthan, A. Jones, and S. Kathikeyan, "Single Phase AC to DC Isolated ZETA Power Factor Correction Converter fed LED drives," *Int. J. Sci. Res. Sci. Eng. Technol.*, pp. 73–80, Jul. 2020, doi: 10.32628/IJSRSET207422.

[133] C. Chaudhary, S. Mishra, and U. Nangia, "A Bridgeless Isolated Zeta Converter for Battery Charging Aided with Solar Photovoltaic System," in *2021 International Conference on Smart Generation Computing, Communication and Networking (SMART GENCON)*, Pune, India: IEEE, Oct. 2021, pp. 1–6. doi: 10.1109/SMARTGENCON51891.2021.9645803.

[134] P. Singh, A. K. Singh, S. S. Kumar, and D. Sharma, "Design and Implementation of Isolated Zeta-Luo Converter for EV Charging Applications," vol. 10, no. 05, 2023.

[135] "[No title found]," *Int. J. Power Electron. Controll. Convert.*.

[136] A. A. Dionizio, L. P. Sampaio, S. A. O. Da Silva, and S. D. J. M. Machado, "Grid-Tied Single-Phase Integrated Zeta Inverter for Photovoltaic Applications," *Energies*, vol. 16, no. 9, p. 3622, Apr. 2023, doi: 10.3390/en16093622.

[137] A. A. Dionizio, G. M. Pelz, L. P. Sampaio, and S. A. O. Da Silva, "Novel Integrated Zeta Inverter for Standalone Applications," *Energies*, vol. 17, no. 11, p. 2748, Jun. 2024, doi: 10.3390/en17112748.

[138] W. Feng, M. Su, L. Jiang, H. Wang, and Y. Liu, "A Common Grounded Zeta-Canonical Switching Cell Single-Stage Buck–Boost Inverter," *IEEE Trans. Power Electron.*, vol. 39, no. 12, pp. 15822–15830, Dec. 2024, doi: 10.1109/TPEL.2024.3443391.

[139] H. R. R and A. J. A, "Fuzzy Logic Control of Zeta Micro Inverter for Solar PV AC Module Fed SPIM," Apr. 06, 2018, *Social Science Research Network, Rochester, NY*: 3160542. Accessed: Jun. 08, 2026. [Online]. Available: https://papers.ssrn.com/abstract=3160542

[140] B. Han, S.-W. Jo, N.-G. Kim, J.-S. Lai, and M. Kim, "Bridgeless Hybrid-Mode Zeta-Based Inverter: Dynamic Modeling and Control," *IEEE Trans. Power Electron.*, vol. 36, no. 6, pp. 7233–7249, Jun. 2021, doi: 10.1109/TPEL.2020.3040113.

[141] M. Khodabandeh and M. Amirabadi, "A Soft-switching Single-stage Zeta-/SEPIC-based Inverter/Rectifier," in *2020 IEEE Applied Power Electronics Conference and Exposition (APEC)*, New Orleans, LA, USA: IEEE, Mar. 2020, pp. 3456–3463. doi: 10.1109/APEC39645.2020.9124509.

[142] K. Tank, M.-M. Garg, N. Gupta, and B. L. Narasimharaju, "Performance Evaluation of Differential Mode Zeta Inverter using Various Modulation Schemes," in *2022 IEEE 2nd International Conference on Sustainable Energy and Future Electric Transportation (SeFeT)*, Hyderabad, India: IEEE, Aug. 2022, pp. 1–6. doi: 10.1109/SeFeT55524.2022.9909249.

[143] I. Shchur, "Bidirectional Single-Stage Zeta-SEPIC DC-AC Converter for Traction BLDC Motors," in *2022 IEEE 3rd KhPI Week on Advanced Technology (KhPIWeek)*, Kharkiv, Ukraine: IEEE, Oct. 2022, pp. 1–6. doi: 10.1109/KhPIWeek57572.2022.9916353.

[144] J. Wijaya and L. H. Pratomo, "Design and Implementation of Current Control for Step Up–Down Zeta H-Bridge Inverter Using STM32F407VET6," *J. INFOTEL*, vol. 16, no. 4, Dec. 2024, doi: 10.20895/infotel.v16i4.1261.

[145] K. Franck, B. H. Zacher, S. Holzmann, M. Wagner, and C. Schumann, "Low THD Current Control of Nonlinear Load Characteristics Using a Single Phase Dual ZETA Inverter," in *2024 IEEE Applied Power Electronics Conference and Exposition (APEC)*, Long Beach, CA, USA: IEEE, Feb. 2024, pp. 2342–2347. doi: 10.1109/APEC48139.2024.10509098.

[146] A. R. M. Guedes, A. A. Dionizio, Ó. P. Westin, L. P. Sampaio, and S. A. O. Da Silva, "Integrated Zeta–Ćuk-Based Single-Phase DC/AC Inverter for Standalone Applications," *Processes*, vol. 13, no. 8, p. 2603, Aug. 2025, doi: 10.3390/pr13082603.

[147] M. Salehi, K. Abbaskhanian, and M. Amirabadi, "A Highly Reliable Zeta-Based AC-Link Universal Converter," *IEEE Trans. Power Electron.*, vol. 40, no. 2, pp. 3210–3224, Feb. 2025, doi: 10.1109/TPEL.2024.3492992.

[148] A. A. Dionizio, L. P. Sampaio, and S. A. O. D. Silva, "Inversor Integrado Zeta para Aplicações em Sistemas Autonômos Monofásicos de Geração de Energia," *Eletrônica Potência*, vol. 27, no. 4, pp. 313–324, Oct. 2022, doi: 10.18618/REP.2022.4.0016.

[149] K. K. H. Dia, M. A. Choudhury, and Ahammad, "A single phase differential Zeta rectifier-inverter," in *2015 IEEE International WIE Conference on Electrical and Computer Engineering (WIECON-ECE)*, Dhaka, Bangladesh: IEEE, Dec. 2015, pp. 284–288. doi: 10.1109/WIECON-ECE.2015.7443919.

[150] L. P. Sampaio, S. Augusto Oliveira Da Silva, and P. J. Silva Costa, "Integrated Zeta Inverter Applied in a Single-Phase Grid-Connected Photovoltaic System," in *2019 IEEE 15th Brazilian Power Electronics Conference and 5th IEEE Southern Power Electronics Conference (COBEP/SPEC)*, Santos, Brazil: IEEE, Dec. 2019, pp. 1–6. doi: 10.1109/COBEP/SPEC44138.2019.9065295.

[151] Ó. P. Westin, R. T. Inuoe, A. A. Dionizio, L. P. Sampaio, and S. A. O. Da Silva, "Improved Modified Zeta Inverter for Single-Phase Grid-Tied System," in *2025 40th Conference on Design of Circuits and Integrated Systems (DCIS)*, Santander, Spain: IEEE, Nov. 2025, pp. 72–77. doi: 10.1109/DCIS67520.2025.11281927.

[152] M. Salehi, K. Abbaskhanian, and M. Amirabadi, "A Highly Reliable Zeta-Based AC-Link Universal Converter," *IEEE Trans. Power Electron.*, vol. 40, no. 2, pp. 3210–3224, Feb. 2025, doi: 10.1109/TPEL.2024.3492992.

[153] M. Salehi, M. Khodabandeh, and M. Amirabadi, "Zeta-Based AC-Link Universal Converter," in *2022 IEEE Energy Conversion Congress and Exposition (ECCE)*, Detroit, MI, USA: IEEE, Oct. 2022, pp. 1–7. doi: 10.1109/ECCE50734.2022.9947744.

[154] L. Li and Q. Zhong, "Novel Zeta-Mode Three-Level AC Direct Converter," *IEEE Trans. Ind. Electron.*, vol. 59, no. 2, pp. 897–903, Feb. 2012, doi: 10.1109/TIE.2011.2161655.

[155] F. L. Tomm, A. R. Seidel, A. Campos, and R. N. Do Prado, "HID lamp electronic ballast based on ZETA AC-AC converter," in *2010 9th IEEE/IAS International Conference on Industry Applications - INDUSCON 2010*, Sao Paulo, Brazil: IEEE, Nov. 2010, pp. 1–6. doi: 10.1109/INDUSCON.2010.5739985.

[156] M. Salehi and M. Amirabadi, "A Soft-Switching Zeta-Based AC-Link Universal Converter," in *2023 IEEE Applied Power Electronics Conference and Exposition (APEC)*, Orlando, FL, USA: IEEE, Mar. 2023, pp. 3233–3239. doi: 10.1109/APEC43580.2023.10131175.

[157] Q. Zhong and L. Li, "Zeta Mode Three-level AC/AC Converter," in *2009 Twenty-Fourth Annual IEEE Applied Power Electronics Conference and Exposition*, Washington, DC, USA: IEEE, Feb. 2009, pp. 1831–1834. doi: 10.1109/APEC.2009.4802919.

[158] Qinglong Zhong and Lei Li, "The non-complementary control strategy of Zeta mode three-level AC/AC converter," in *2009 IEEE 6th International Power Electronics and Motion Control Conference*, Wuhan: IEEE, May 2009, pp. 1015–1018. doi: 10.1109/IPEMC.2009.5157533.

[159] L. Lei and Q. Zhong, "The output spectrum of zeta mode three-level AC/AC converter," in *2012 7th IEEE Conference on Industrial Electronics and Applications (ICIEA)*, Singapore: IEEE, Jul. 2012, pp. 628–631. doi: 10.1109/ICIEA.2012.6360802.

[160] N. Jeelani and A. H. Bhat, "Single-Phase AC-AC Zeta Converter-Based Automatic AC Voltage Regulator," in *2022 1st International Conference on Sustainable Technology for Power and Energy Systems (STPES)*, SRINAGAR, India: IEEE, Jul. 2022, pp. 1–6. doi: 10.1109/STPES54845.2022.10006612.

[161] M. Salehi and M. Amirabadi, "Zeta-Based Universal Converter with Zero-Current Switching," in *2023 IEEE Energy Conversion Congress and Exposition (ECCE)*, Nashville, TN, USA: IEEE, Oct. 2023, pp. 2438–2444. doi: 10.1109/ECCE53617.2023.10362036.